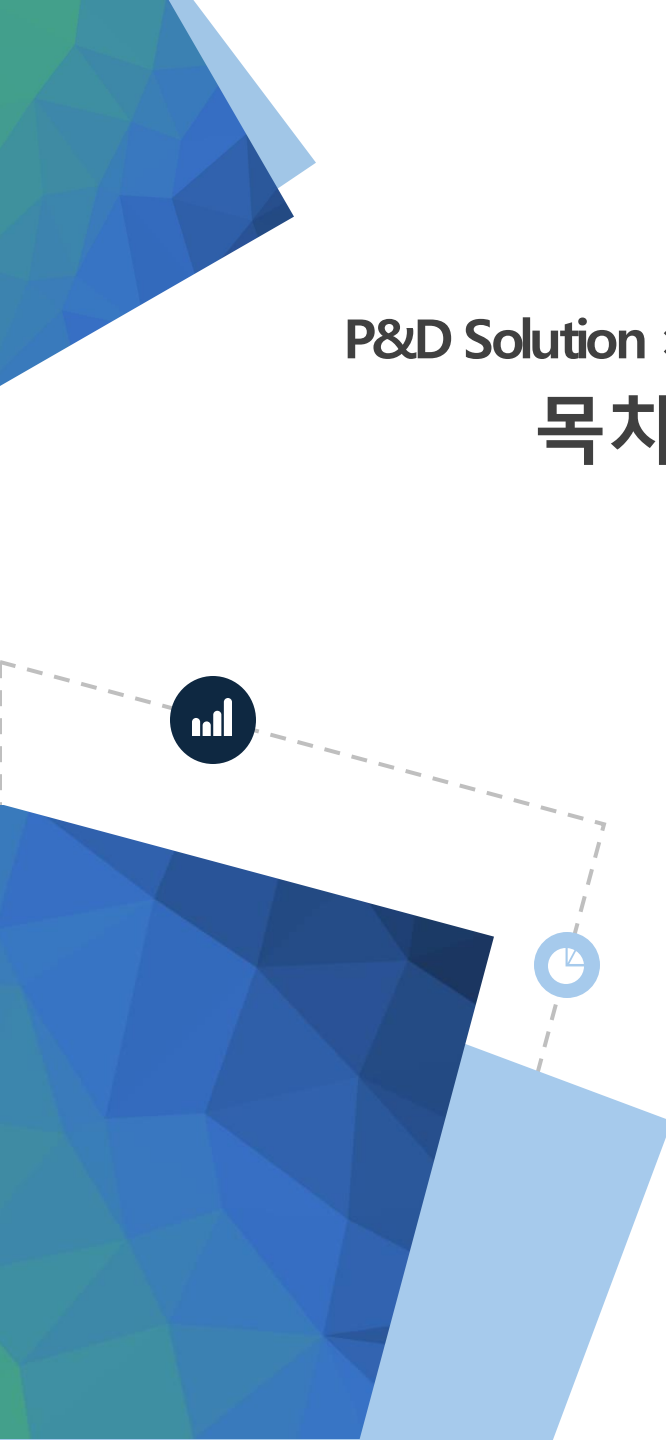


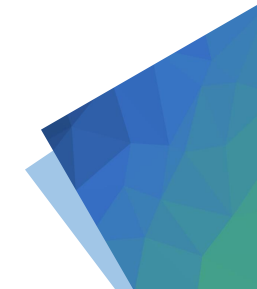
Dataiku – 관리자 교육





P&D Solution × Dataiku

목차

1. Dataiku 개념 및 아키텍처
 2. 설치 및 초기 설정
 3. Connection 관리
 4. 사용자 및 권한 관리
 5. 시스템 운영 관리
 6. 확장 및 고급 설정
- 



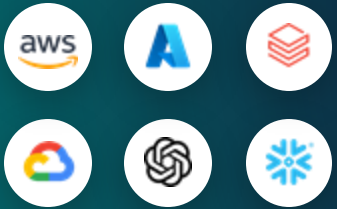
1. Dataiku DSS 아키텍처

Dataiku는 통합된 “Universal AI Platform” 입니다



Your Teams / Users

분석가와 실사용자, 코드 개발자와
클릭 GUI 기반 사용자를 위한 모든
기능 제공



Your Technology Optionality

거대언어모델 (LLMs), Data Platform 및
Cloud 기술을 모두 사용할 수 있는 옵션은
현재와 미래 사용자를 위한 권리



Your Data

모든 유형의 데이터, 모든 컴퓨팅
환경을
현재 환경 그대로 활용



Your Governance

데이터 전처리, 모델링, 운영(MLOps)을
포함하여
데이터에서 AI까지 라이프사이클의 모든
단계를 완전히 관리 감독



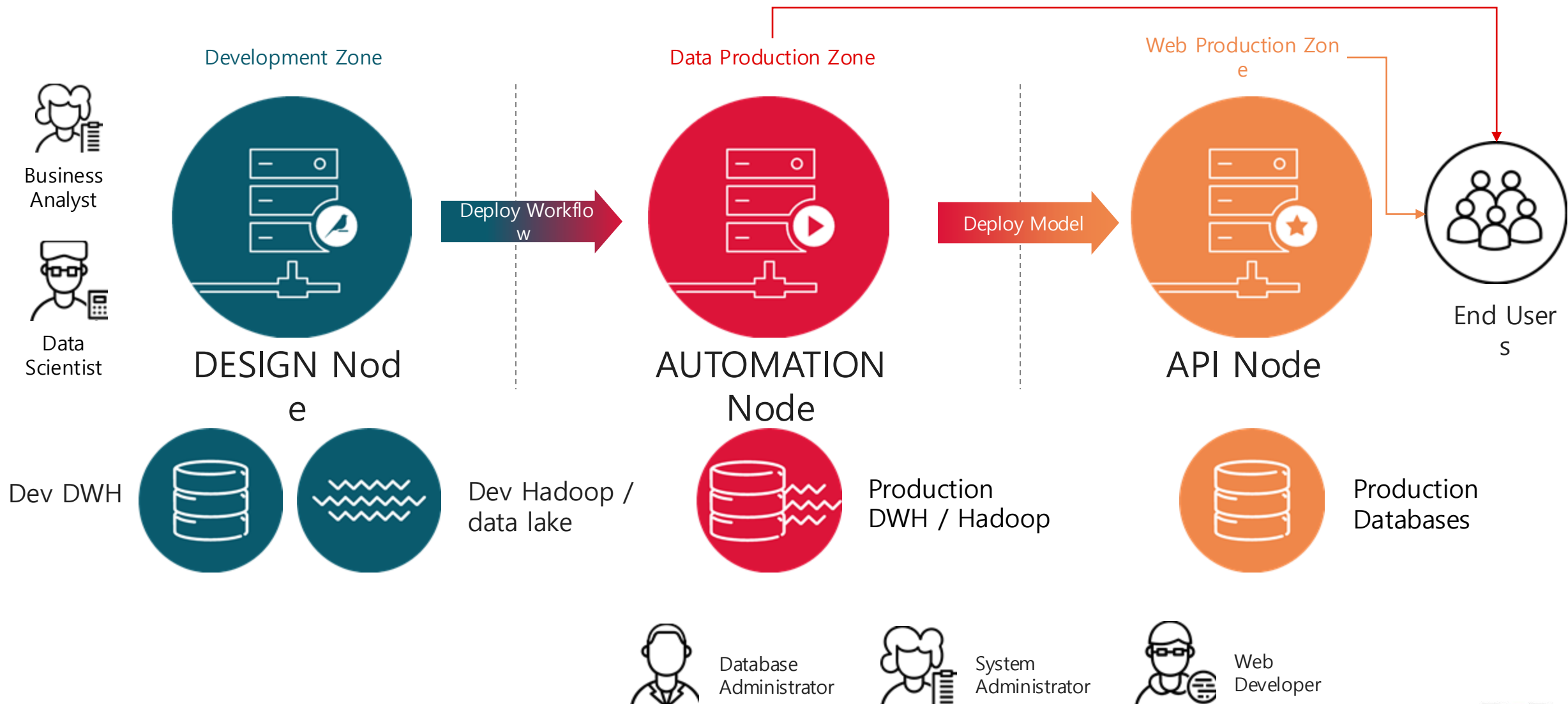
**YOUR AI
STRATEGY**

Everyday AI for Everyone

모든 공정을 하나의 플랫폼에서 실행하여 효율적인 AI 환경을 구축

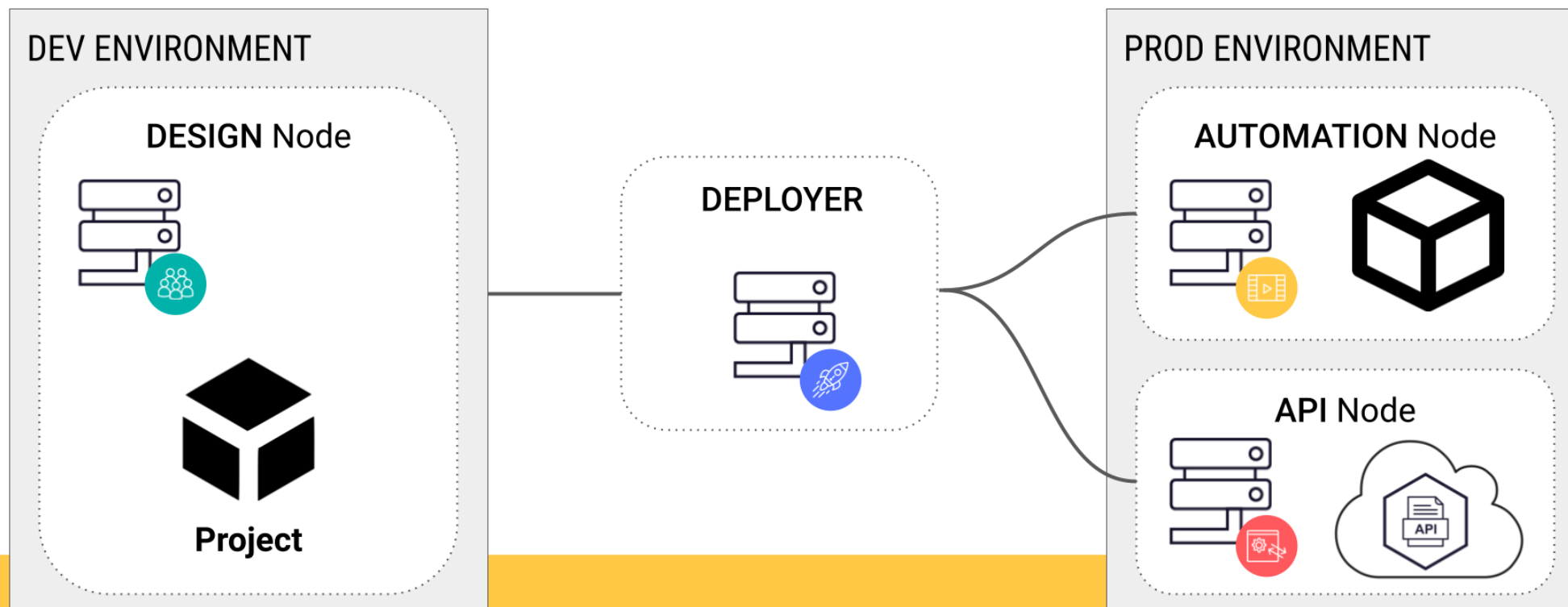


Dataiku DSS 아키텍처

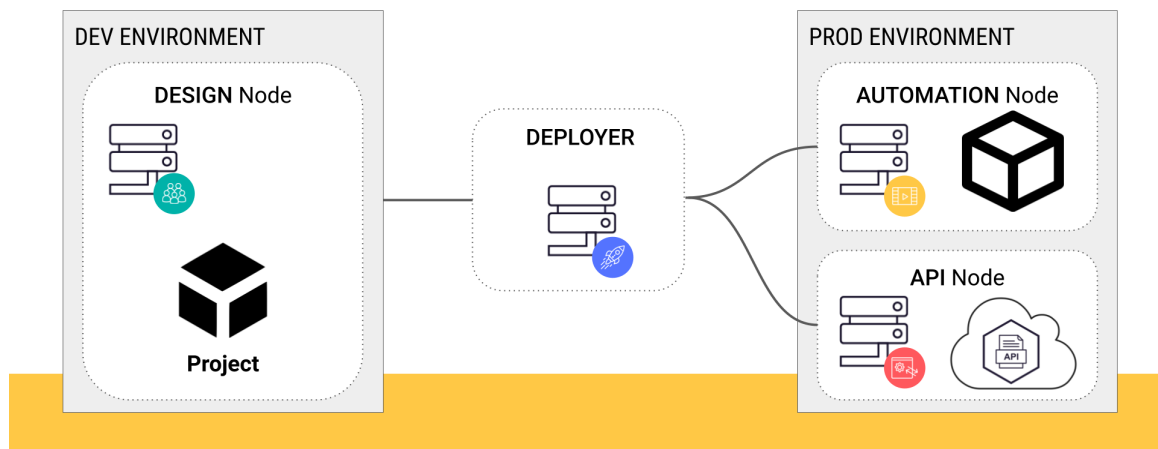




DESIGN AND OPERATIONALIZE



DESIGN AND OPERATIONALIZE



Design Node

- **용도:** 개발·실험 환경
- 데이터 탐색, 전처리, 모델 학습, Flow 구성 등 **모든 작업을 수행하는 메인 작업 공간**
- 데이터 사이언티스트와 분석가가 프로젝트를 만들고 실험하는 곳
- 인터랙티브하게 코드 실행, 노트북 사용, 시각화 등 가능
- 쉽게 말해 "**개발 서버**"

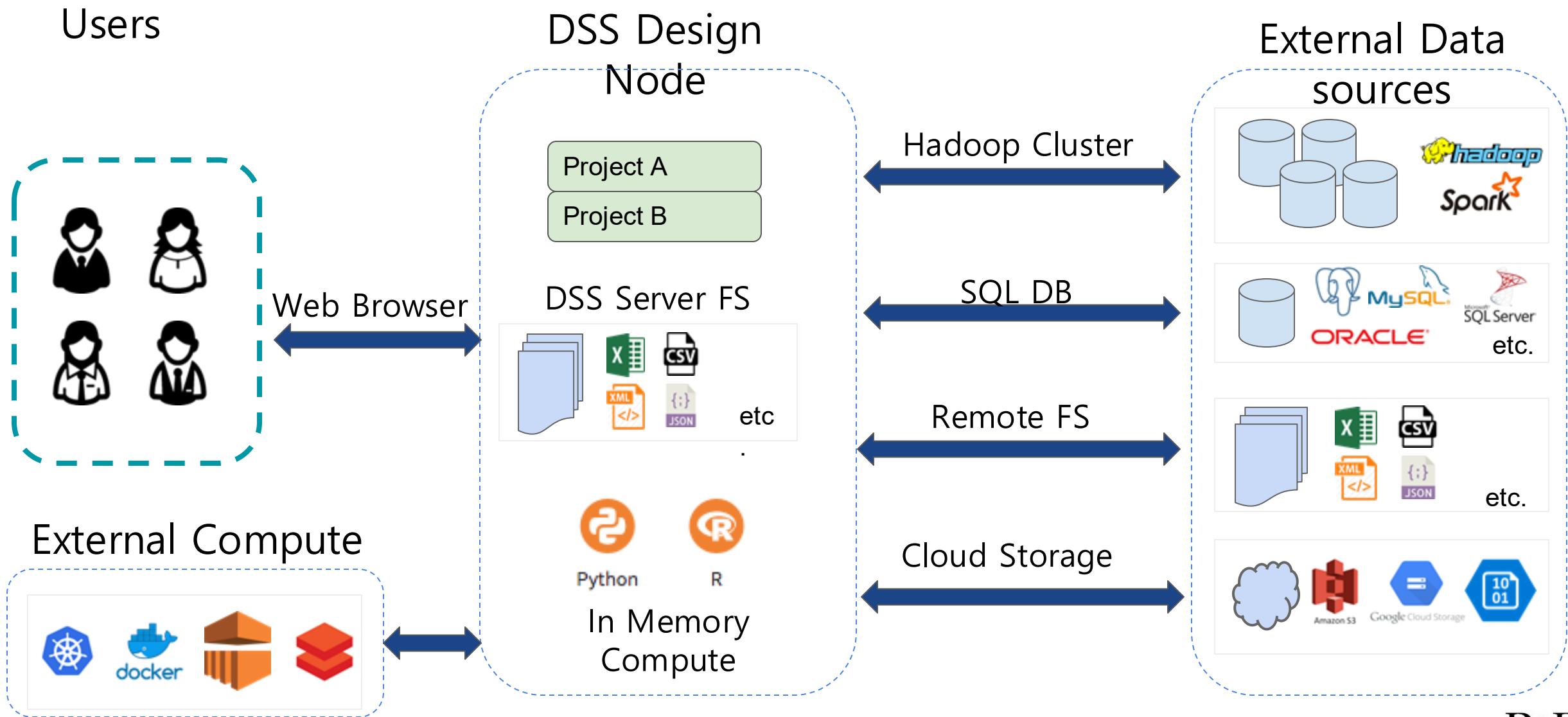
Automation Node

- **용도:** 운영·스케줄링 환경
- Design Node에서 완성된 프로젝트를 패키징해서 배포
- 스케줄러로 파이프라인을 **정기 실행** (배치 처리)
- 시나리오 기반 자동화를 운영
- 사용자가 직접 Flow를 수정하거나 실험하지 않음
- 쉽게 말해 "**운영/프로덕션 서버**"

API Node

- **용도:** 실시간 서빙 환경
- 학습된 모델을 **REST API 엔드포인트**로 노출하여 실시간 예측(inference) 제공
- 외부 애플리케이션에서 HTTP 요청으로 모델 호출 가능
- 로드밸런싱, 멀티 엔드포인트 관리 지원
- 쉽게 말해 "**API 서빙 서버**"

Dataiku DSS 아키텍처 (simplified)

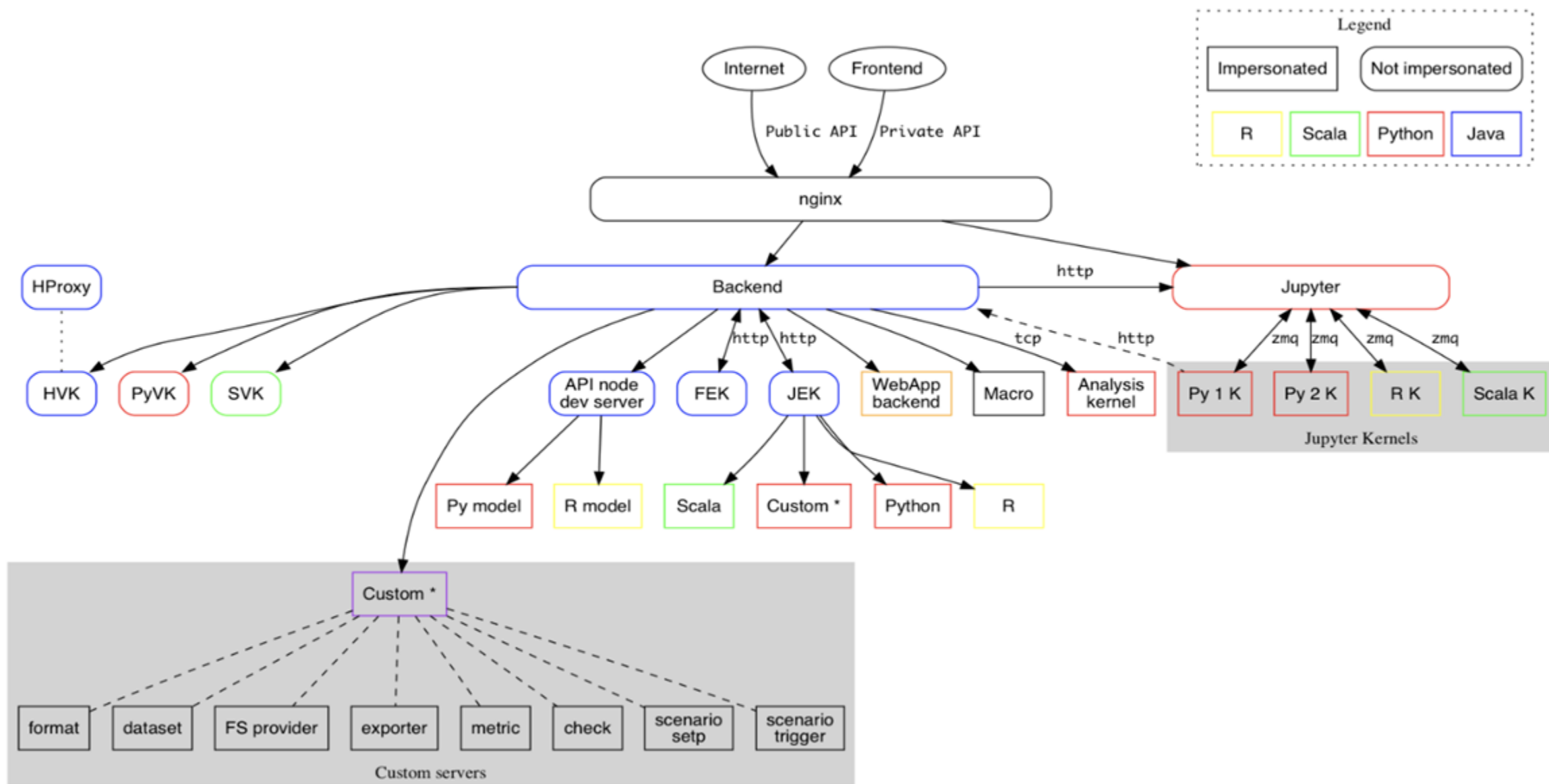


```
dataiku@pndiku:/data/dataiku/design$ ./bin/dss start
Waiting for DSS supervisor to start ...
→ backend                STARTING
→ ipython                 STARTING
→ nginx                   STARTING
DSS started, pid=7498
Waiting for DSS backend to start
dataiku@pndiku:/data/dataiku/design$
```

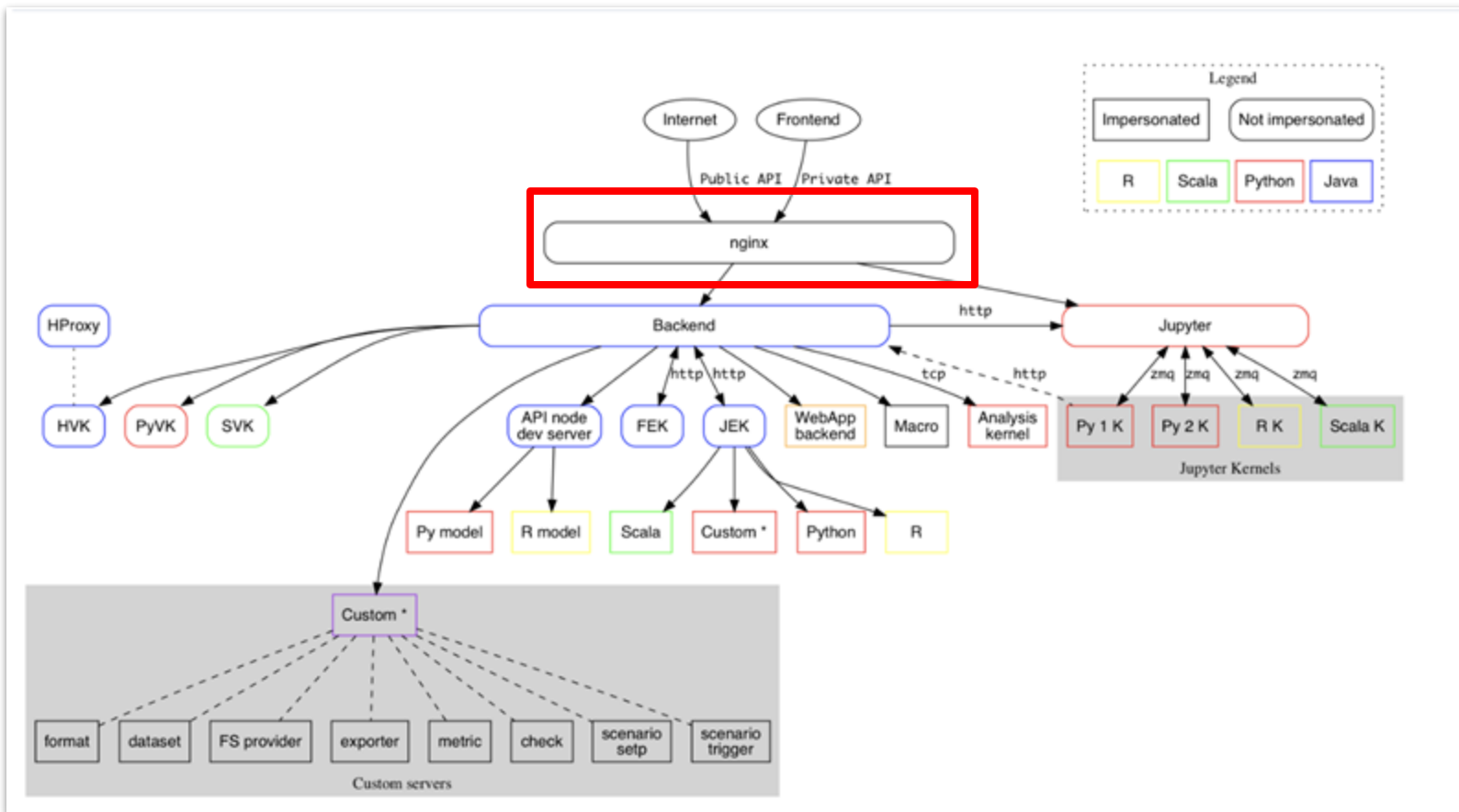
DSS Design/Automation Node 기동하면

- 4개 프로세스가 포크 됨
 - Supervisor: process manager로 아래 3개의 top-level 프로세스를 관리
 - Nginx server listening to installation port
 - Backend server listening to installation port + 1
 - Ipython (Jupyter) server listening to installation port + 2

DSS 컴포넌트와 프로세스 > Core 아키텍처



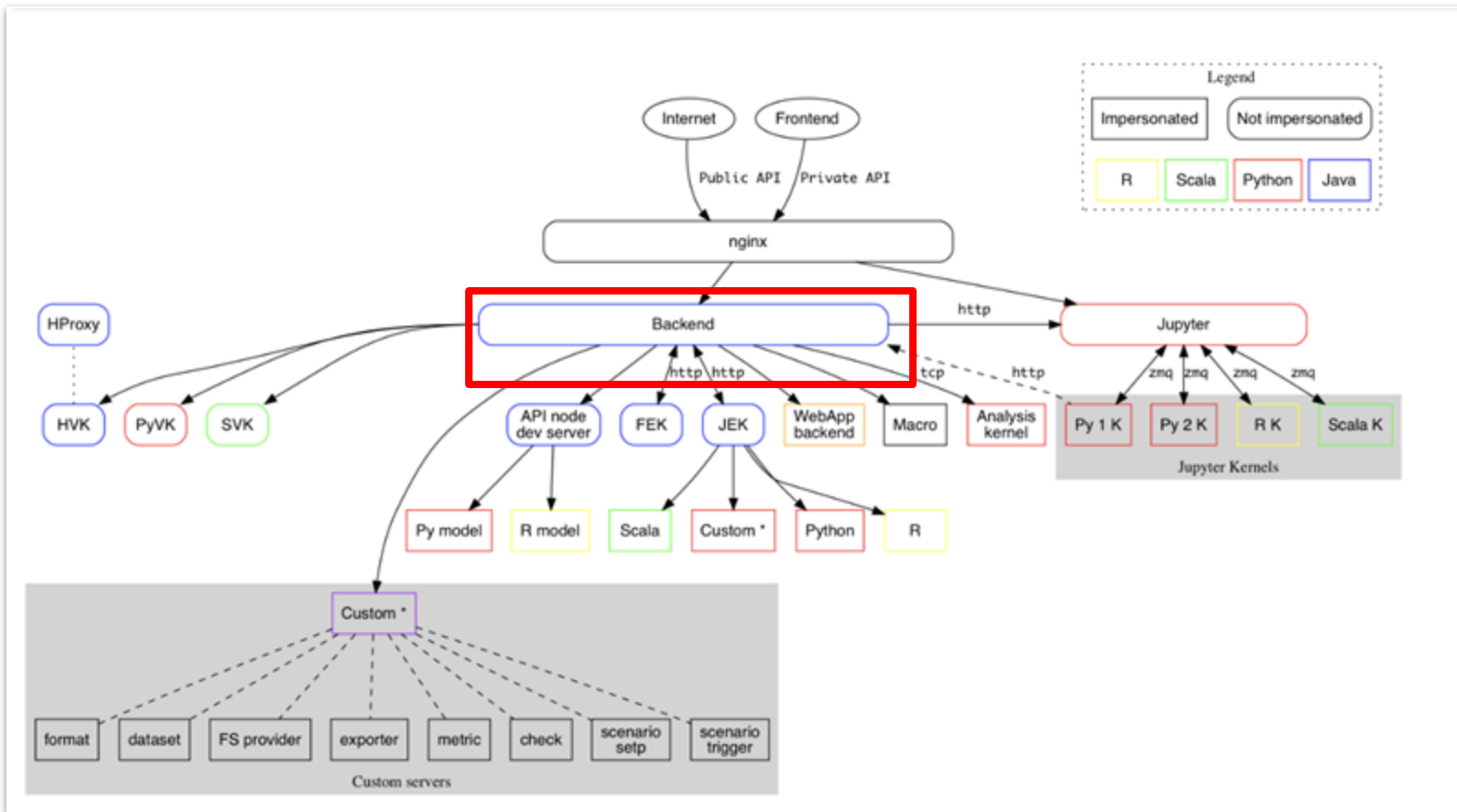
DSS 컴포넌트와 프로세스 > nginx



NGINX

- 웹 브라우저를 통한 사용자의 모든 요청을 처리 함
- 다른 DSS 컴포넌트로 요청을 할 때 HTTP Proxy 역할을 수행 함
- 설치 시 지정한 port를 사용 함
- Protocol: HTTP(s) and websockets.

DSS 컴포넌트와 프로세스 > backend



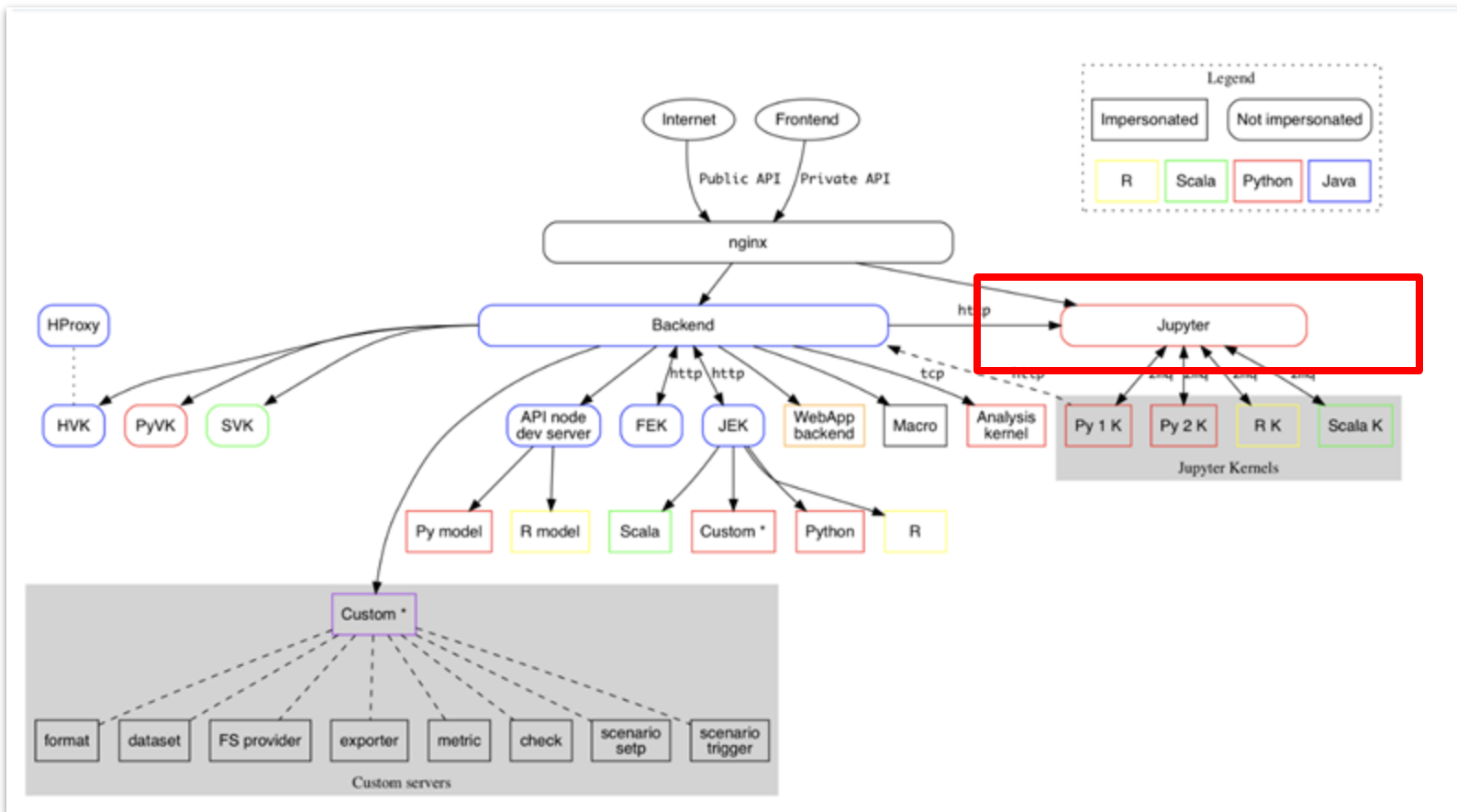
BACKEND

작업·자원·권한을 조율하는 두뇌
(Orchestrator)

- config 폴더 접근
 - 전처리 데이터 미리보기
 - 데이터셋 탐색(집계,통계,차트계산)
 - git 연동
 - public api 대응
 - 스케줄 처리
 - 시나리오 수행
-
- DSS port +1 사용

- Backend 하위 프로세스 실행
- 시나리오/스케줄러로 잡을 큐잉·모니터링하고 로그/상태를 기록

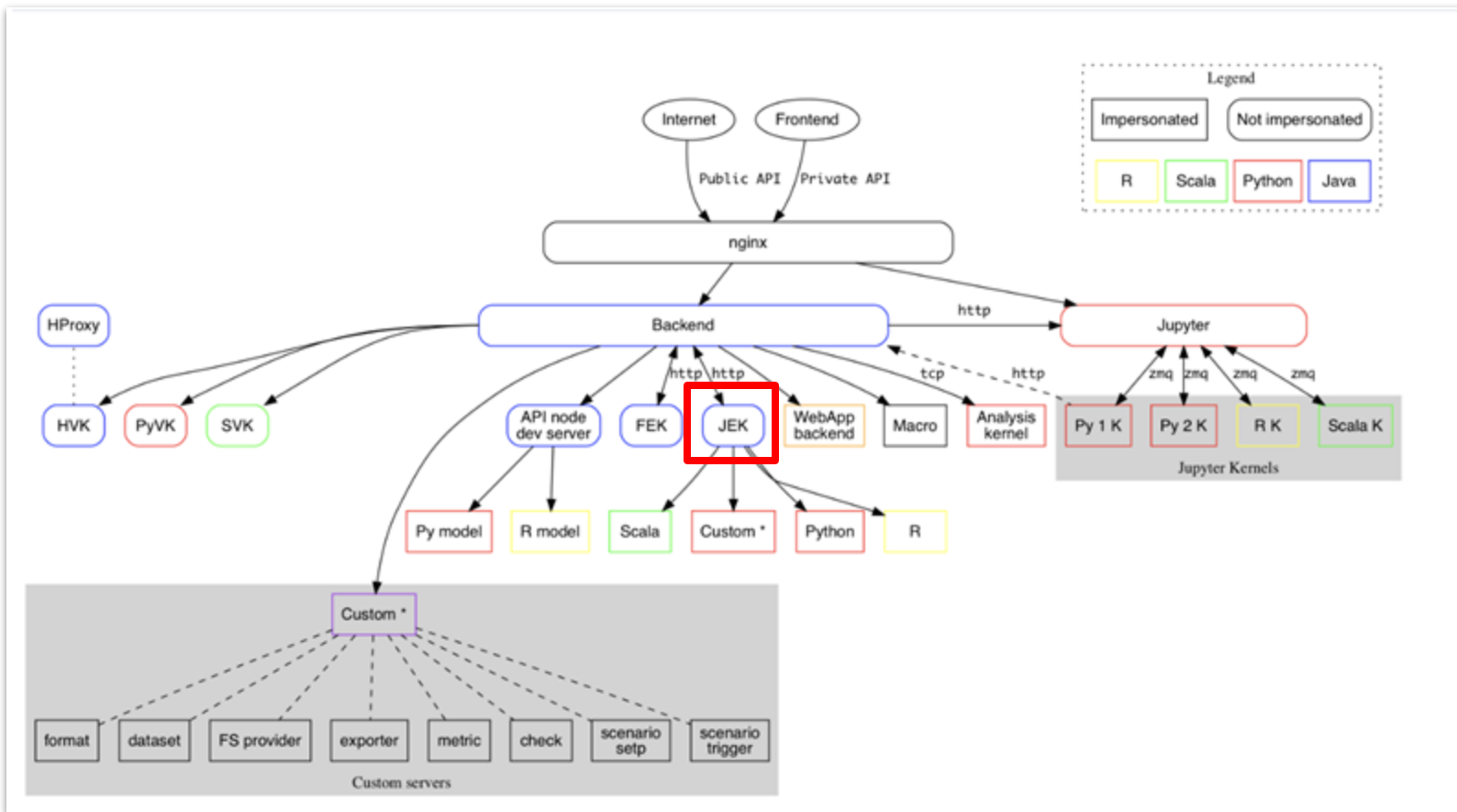
DSS 컴포넌트와 프로세스 > Ipython(Jupyter)



IPYTHON (JUPYTER)

- Python, R, Scala 노트북 커널과 통신 처리 수행
- ZMQ 프로토콜
- DSS port + 2 사용

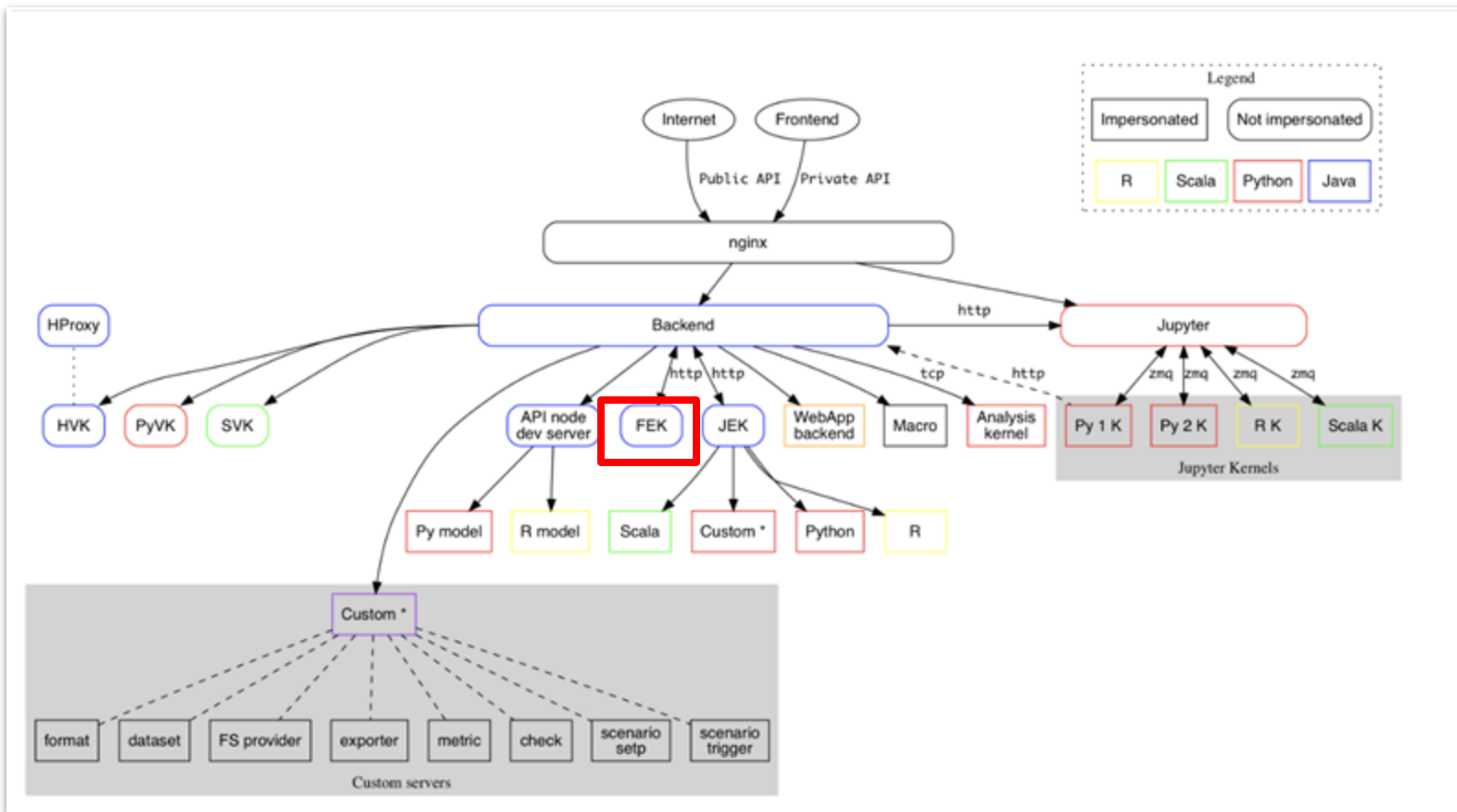
DSS 컴포넌트와 프로세스 > JEK



JOB EXECUTION KERNEL (JEK)

- **핵심역할:** 의존성 계산 및 **DSS 엔진**에서 레시피 실행
- 타 엔진/코드 실행: Python, R, Spark, SQL 등
- 이때, 자식 프로세스 기동

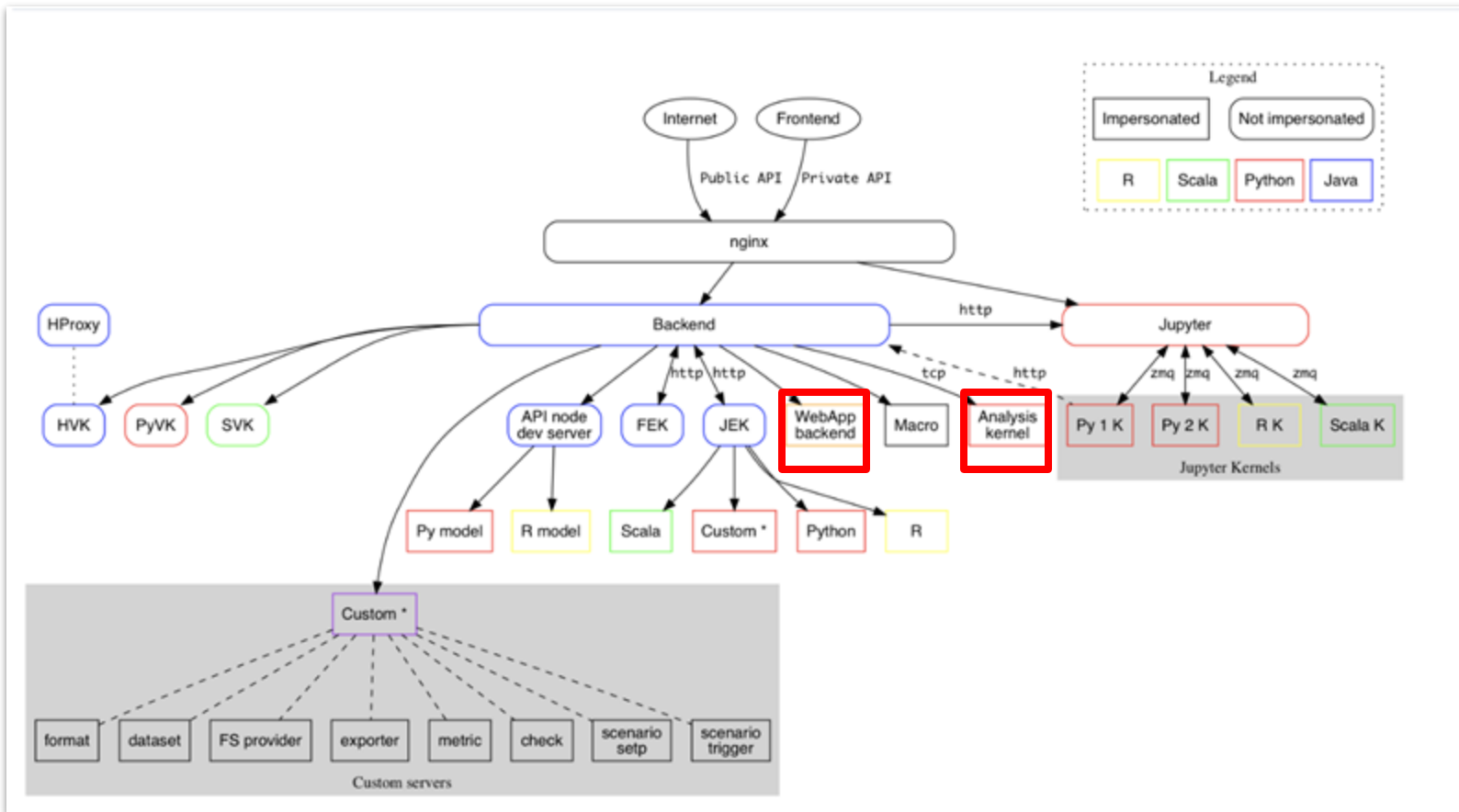
DSS 컴포넌트와 프로세스 > FEK



FUTURE EXECUTION KERNEL(FEK)

- **FEK 역할:** Job이 아닌 백그라운드 위험 작업 전담
 - 메트릭 계산
 - ML 학습/차트용 데이터 샘플 빌드
 - ML 준비단계: 전처리/특성화 등 학습 준비 작업 수행

DSS 컴포넌트와 프로세스 > 기타



ANALISIS SERVER

- Python 기반의 머신러닝 학습과 데이터 전처리 수행

WEBAPP BACKEND

- 사용자가 작성한 WebApp 백엔드 실행
 - Python Flask Backend
 - Python Bokeh
 - R Shiny

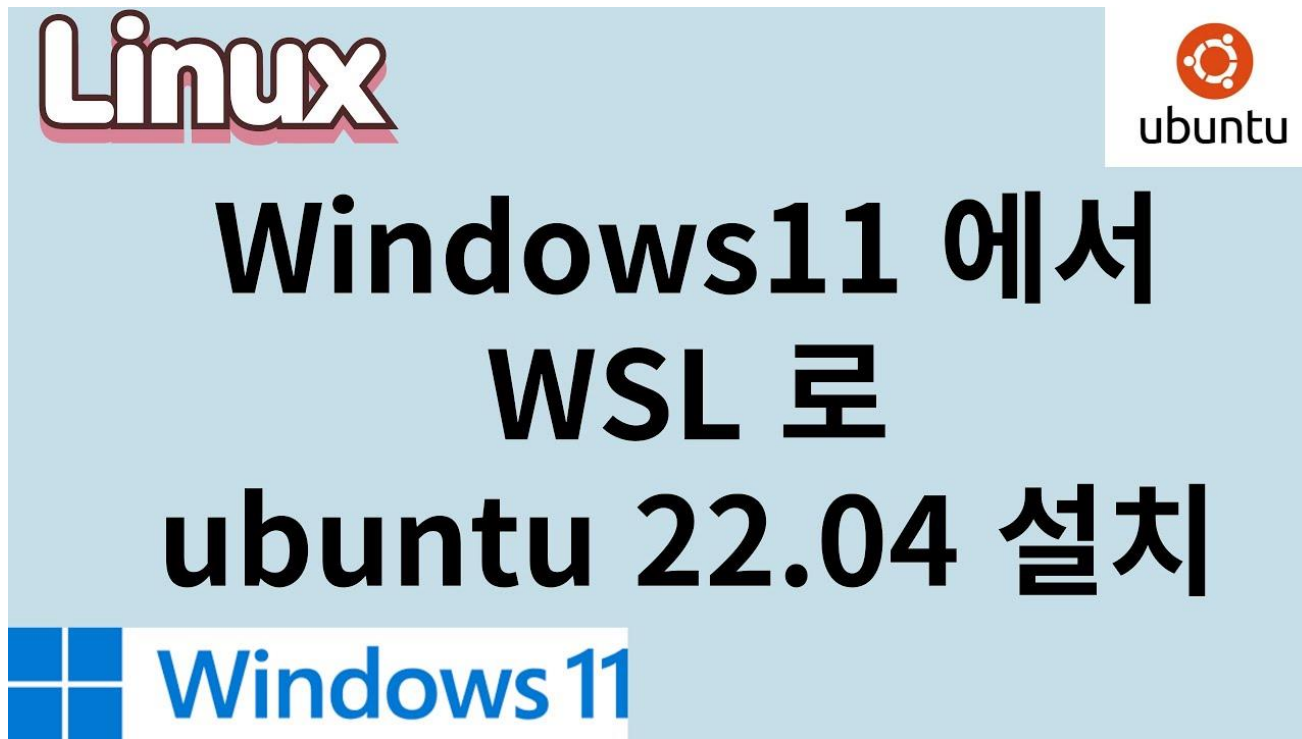


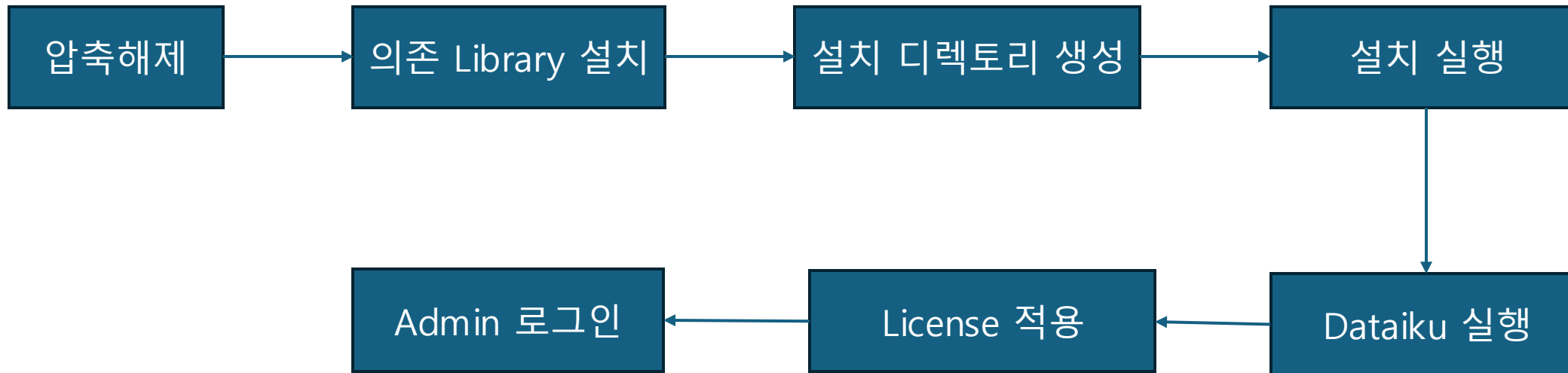
2. Dataiku 설치 및 초기 설정

설치 > 사전 준비 사항

❖ 사전 준비 사항

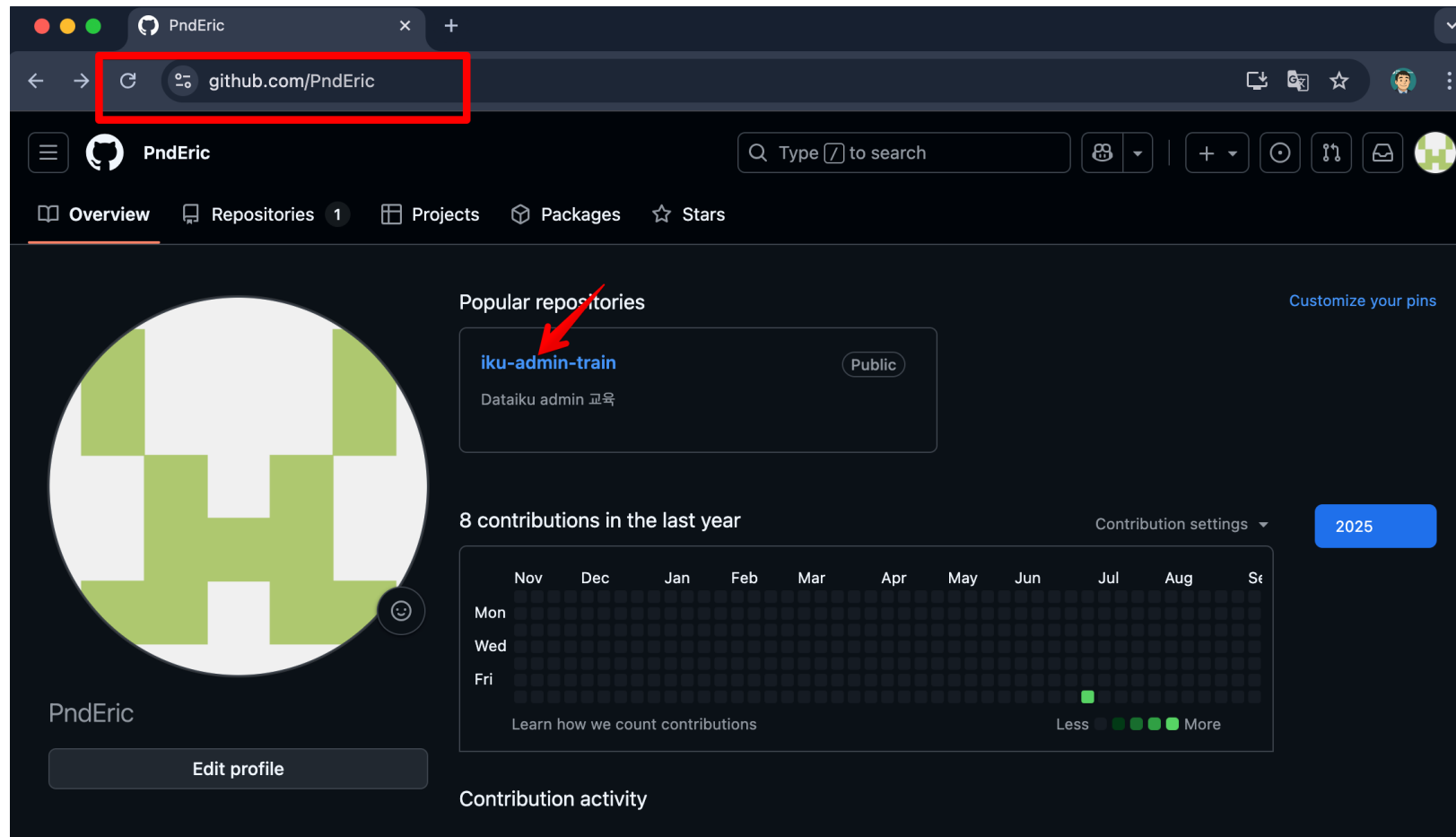
- 1) WSL2 on Windows OS
- 2) Dataiku 설치 압축 파일 download





Node Type	Port number	Directory
Design	10010	/data/dataiku/design
Automation	10020	/data/dataiku/automation
API	10030	/data/dataiku/api

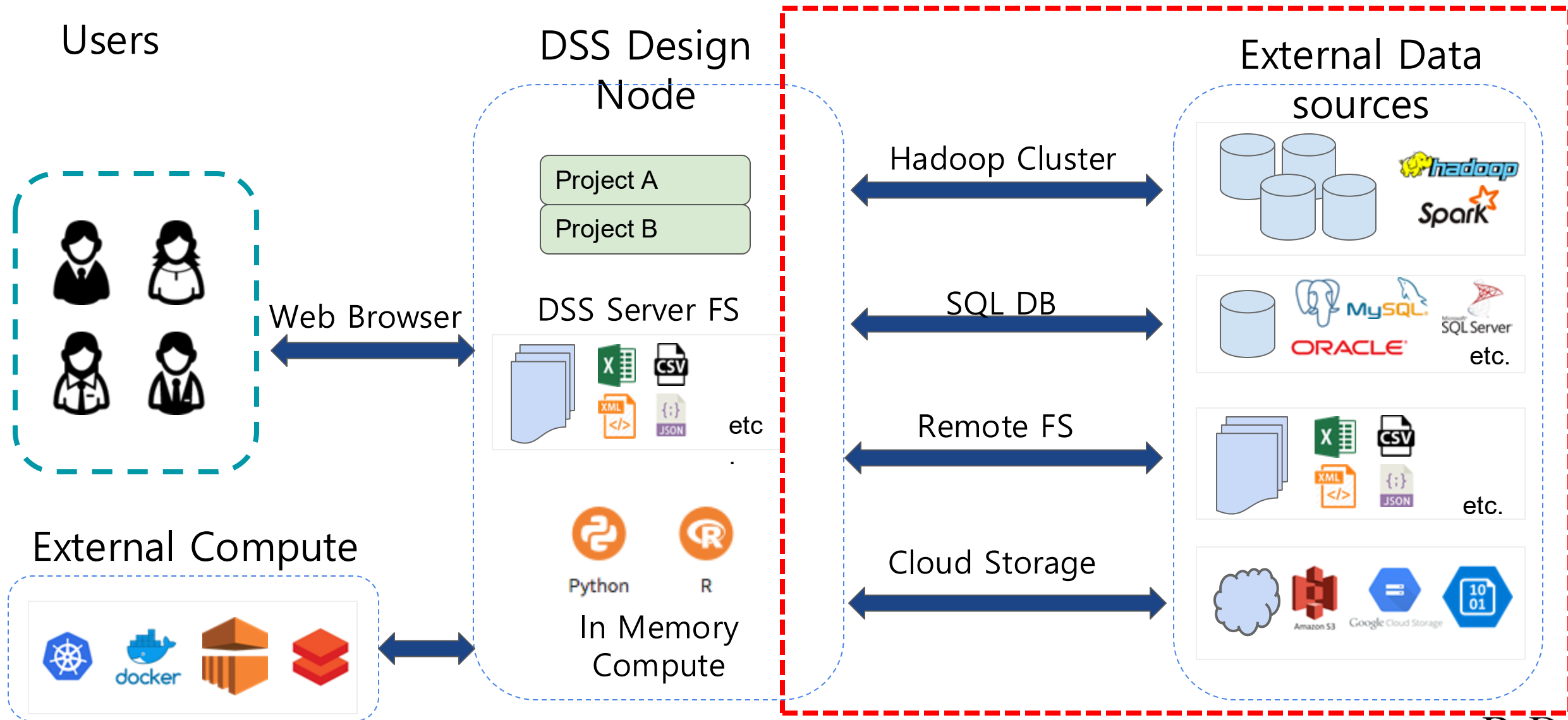
<https://github.com/PndEric>

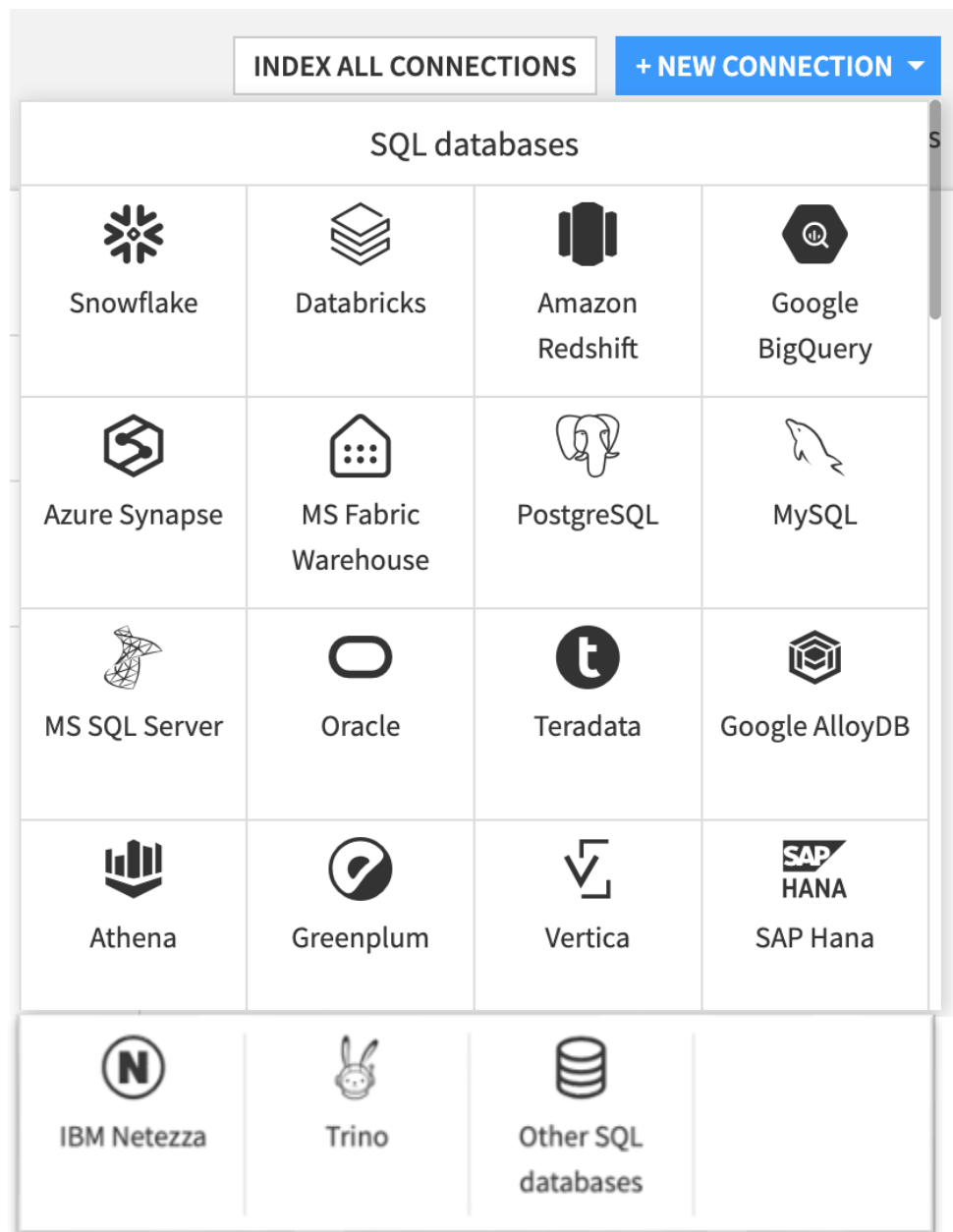




3. Connection 관리

Dataiku DSS 아키텍처

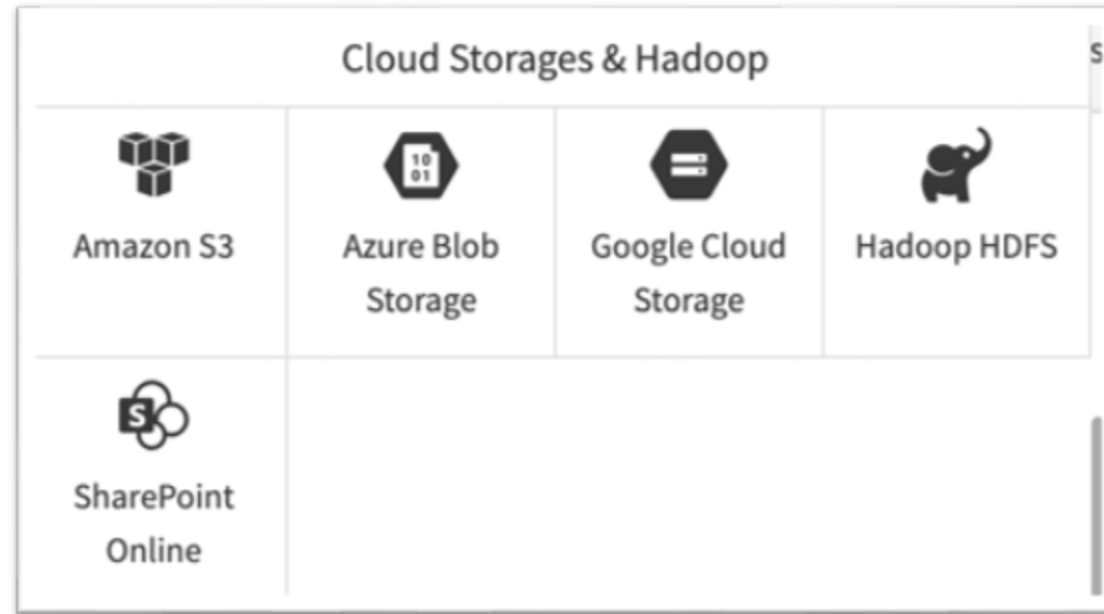




NoSQL

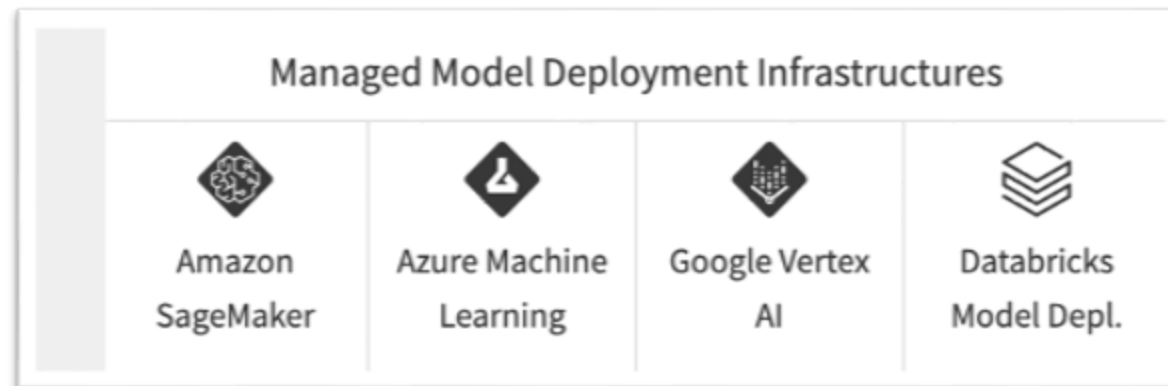
NoSQL			
 MongoDB	 Cassandra	 ElasticSearch	

Cloud Storages & Hadoop



LLM Mesh

LLM Mesh			
 OpenAI	 Azure OpenAI	 Azure LLM	 Cohere
 Anthropic	 Vertex Generative AI	 AWS Bedrock	 Databricks Mosaic AI
 Snowflake Cortex	 MistralAI	 Local Hugging Face	 Amazon SageMaker LLM
 Stability AI	 Custom LLM	 Pinecone	 Azure AI Search



Lab. SQL DB Connection 설정 Flow

Dataiku DSS 웹 UI를 통해서 SQL DB 연결을 생성 함

- Administration > Connections
- “New connections” 클릭 후 연결 할 DB 유형 선택
- Connection 입력
- IP, Port, 계정 등 연결에 필요 한 정보 입력
- 화면 하단 Test 버튼 클릭
- Connection 정보 저장

Lab. PostgreSQL 연결

Dataiku DSS

Search DSS...

License Connections Security Clusters Code Studios Code Envs Monitoring Maintenance Settings

DSS settings

SQL databases

Snowflake Databricks Amazon Redshift Google BigQuery

Azure Synapse MS Fabric Warehouse **PostgreSQL** MySQL

MS SQL Server Oracle Teradata Google AlloyDB

Athena Greenplum Vertica SAP HANA

IBM Netezza Trino Other SQL databases

New PostgreSQL connection

New connection name: Training-Postgres

Basic params

Host: 175.212.202.76

Database: dataiku

Port: 15432

User: dataiku

Password:

Advanced JDBC properties

+ ADD PROPERTY

Use custom JDBC URL: ☐ Used to customize JDBC connection URL

Advanced params

Enable PostGIS support: ☐ PostGIS extension should be installed in a PostgreSQL database

Driver to use: Dataiku-managed

Send driver jars to Spark: ☐ Experimental: Add driver jars to spark-submit subprocesses

Fetch size: Emotv = default

Naming rules for new datasets

These settings define how managed SQL datasets are mapped to database tables. These settings are applied when creating a new managed SQL dataset. You can always modify these afterwards in the SQL dataset settings. They are also used to create temporary tables if the connection is used for bundled data. See [the documentation](#) for more information.

Schema

Managed datasets will be created in this schema. You may reference variables here, they will be expanded dynamically

Allow schema override

☐ If this is enabled, an option to choose a schema will be presented to users each time they create a managed dataset on this connection

Table prefix

Table name for new managed datasets will be prefixed by this. You may reference variables here, they will be expanded dynamically

Table suffix

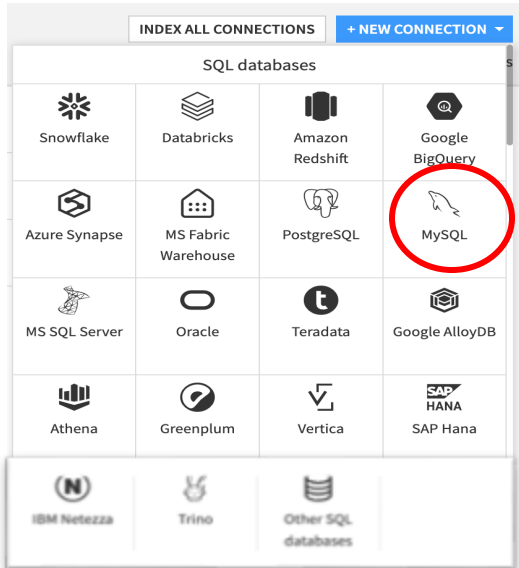
Table name for new managed datasets will be suffixed by this. You may reference variables here, they will be expanded dynamically

TEST **CREATE**

Connection OK

Lab. MySQL 연결(1/2)

[제품 문서 링크](#)



예상결과: fail

MySQL connection: Training-MySQL

Basic params

Host	175.212.202.76	
Database	dataiku	Mandatory
Port	13306	
User	dataiku	
Password		
Advanced JDBC properties	allowPublicKeyRetrieval → true	☐ Secret
	useSSL → false	☐ Secret
	+ ADD PROPERTY	

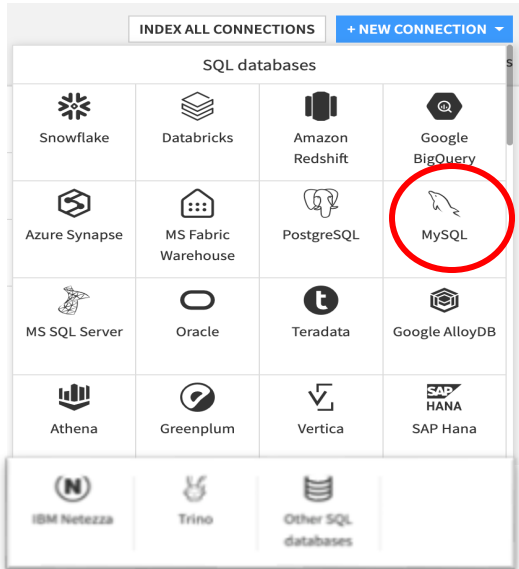
Use custom JDBC URL ☐ Used to customize JDBC connection URL

Advanced params

Database version	>=8.0		
Driver jars directory	 Path to a directory containing the JDBC driver jars. The path can be absolute or relative to the DSS data directory. If empty, jars must be copied in [DSS home]/lib/jdbc dir		
Send driver jars to Spark	☐ Experimental: Add all jars of [DSS home]/lib/jdbc to spark-submit subprocesses. Use a Jars directory to pass only this connections's driver.		
Post-connect commands	These statements are run after the connection is created <table><tr><td>1</td><td> </td></tr></table>	1	
1			
Job post-connect commands	When used in a job context, these additional statements are run after the connection is created <table><tr><td>1</td><td> </td></tr></table>	1	
1			

TEST SAVED !

Connection OK



- 1) MySQL [JDBC driver](#) 다운로드.
- 2) Stop DSS: `./DATA_DIR/bin/dss stop`
- 3) JDBC driver JAR 복사 → `DATA_DIR/lib/jdbc` directory
- 4) Start DSS: `./DATA_DIR/bin/dss start`

→ git hub 실습자료#2 참조

SQL Connection 파라미터(1/3)

The screenshot shows a configuration window for a SQL connection. It is divided into several sections:

- Advanced JDBC properties**: Includes a checkbox for "Use custom JDBC URL" and a "+ ADD PROPERTY" button.
- Advanced params**: Includes checkboxes for "Enable PostGIS support", "Driver to use" (set to "Dataiku-managed"), "Send driver jars to Spark", and "Fetch size" (with a text input field and "Empty = default" note).
- Schema search path**: Includes a text input field and a note "This search path will be set on all sessions. Useful for SQL queries and notebooks".
- Post-connect commands**: Includes a text area for commands to run after connection creation.
- Job post-connect commands**: Includes a text area for commands to run after connection creation in a job context.
- Truncate to clear data**: Includes a checkbox for "When the columns did not change, should DSS truncate tables instead of dropping them?".
- Indexing**: Includes checkboxes for "Index indices", "Index foreign keys", and "Index system tables".

Advanced JDBC properties

- 모든 DB에서 임의의 키/값 속성을 JDBC 드라이버로 그대로 전달 가능
- 사용 가능한 속성은 각 JDBC 드라이버마다 다름 → 해당 드라이버 문서 참고
- ex) sslmode=true, reWriteBatchedInters=true

Fetch size

- DB에서 레코드를 배치 단위로 가져와 성능 향상
- Fetch size로 배치 크기 지정 가능

SQL Connection 파라미터(2/3)

Naming rules for new datasets

These settings define how managed SQL datasets are mapped to database tables. These settings are applied when creating a new managed SQL dataset. You can always modify these afterwards in the SQL dataset settings. They are also used to create temporary tables if the connection is used for bundled data. See [the documentation](#) for more information.

Schema 
Managed datasets will be created in this schema. You may reference variables here, they will be expanded dynamically

Allow schema override ☐
If this is enabled, an option to choose a schema will be presented to users each time they create a managed dataset on this connection

Table prefix 
Table name for new managed datasets will be prefixed by this. You may reference variables here, they will be expanded dynamically

Table suffix 
Table name for new managed datasets will be suffixed by this. You may reference variables here, they will be expanded dynamically

Date settings

Default assumed timezone
Default "assumed time zone" selection when adding a dataset through this connection. This will be the default timezone assigned to timezone-less data when read in as a DSS datetime with tz.

Custom properties

Advanced connection properties [+ ADD PROPERTY](#)
For specific use cases

변수를 활용한 룰 정의

- SQL database 세팅에서 변수를 활용하여 Table 이름, 스키마에 대한 규칙을 정의할 수 있음
- 예시)
 - Schema: DATAKU
 - Table name prefix: \${projectKey}_
 - Table name suffix: \${myvar1}

SQL Connection 파라미터(3/3)

Usage params

Allow write

☒ Can DSS write to the datasets from this connection?

Allow managed datasets

☒ Can new managed datasets be created on this connection?

Max nb. of activities

Limit the number of concurrent activities involving this connection.

Security settings

See [the documentation](#) for more information.

Freely usable by

☒ Every analyst
☐ Selected groups
Who can create new datasets in this connection, and more generally "browse" this connection.

Details readable by

☒ Nobody
☐ Every analyst
☐ Selected groups
Who can access the details of the connection (including credentials, if connection has some).

사용 제약

- 이 연결을 통한 Read/Write 허용 여부 지정 가능
- 이 연결을 누가 사용 할 수 있는지 제약도 가능 함

SQL Connection > 사용자 별 인증

Credentials

See [the documentation](#) for more information.

Credentials mode

Per user

Custom provider (advanced)

Class name of a custom basic credential provider

사용자 별 인증

- DB 연결 시 보안 요건으로 개별 사용자별로 인증이 필요한 경우가 있음
- Credential mode를 Global 이 아닌 “Per user” 모드 사용

Dataiku DSS

Search DSS...

홍길동

My Settings

Exports

Stars and watches

API keys

Credentials

My files

My Account

LOGOUT

G

Login: @gildong

Profile: FULL_DESIGNER

Groups:

- AI_SECTION
- data_team

Personal credentials

Connections

Connect your account to your external data through the Dataiku platform. You can either insert ID/PASSWORD right away, or connect to the 3rd party right away depending on the setup by admin.

If you have questions, see the documentation or contact your administrator.

Connection type	Connection name	Type	Status	Actions	Last update
PostgreSQL	PerUserConn	Password	Filled (gildong)	UPDATE REMOVE	Today at 12:50

Plugins

Plugins add extra features (e.g., data visualization) to enhance what you can do within Dataiku DSS.

If you have questions, see the documentation or contact your administrator.

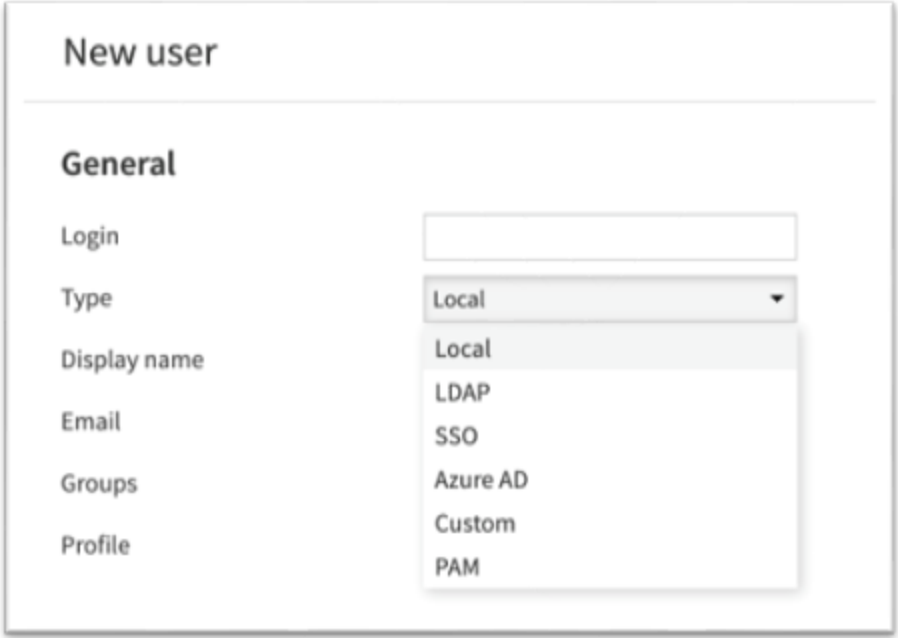


4. 사용자 및 권한 관리

User Identity & Authentication

- 사용자는 다음 중 1곳에서 식별 됨:

- local db
- LDAP/AD
- SSO
- Azure AD
- Custom
- PAM



The screenshot shows a 'New user' form with a 'General' tab. The 'Type' dropdown menu is open, showing options: Local, LDAP, SSO, Azure AD, Custom, and PAM. The 'Login' field is empty, and the 'Display name' field is also empty. The 'Email' field is empty. The 'Groups' field is empty. The 'Profile' field is empty.

- 참고) MFA 옵션

- 권장: DSS로그인을 SSO로 위임하고 IdP(*Okta, Azure AD*등) 에서 MFA 정책 적용
- PAM 선택 후 서버 OS의 PAM 모듈에서 2FA 추가

Permission Model

- User:
 - DSS login 인증에 존재해야 함
 - 사용자 Group에 속함
 - Profile을 가짐
- User Profile:
 - 라이선스 적용을 위한 개념

Permission Model

- Group:
 - 여러 User들의 묶음
 - RBAC? No, Dataiku use "GBAC"
 - Global Permission을 정의
 - are you admin? Can you create connections?
- Projects:
 - 각 Ggroup에 권한 설정
 - Project레벨의 세팅을 강제 할 수 있음
- Data Connections:
 - 각 Ggroup에 access 권한 부여
 - 특정 connection은 사용자별 인증 허용 함

Permission Model > Users

- User:profile+group 권한 따름
 - 사용자 설정에 의해 자동으로 결정 됨

Edit user "eric"

General

Type

Local

Display name

조상철

Email

sccho@pndsolution.com

Groups

administrators, data_team

Profile

Full Designer

- Auth Matrix: 모든 프로젝트에 대한 사용자 별 접근여부와 권한을 확인할 수 있음

By user	SELECT USERS	조상철 @eric	테스트용 사용자 @testuser	pnd00 @pnd00
Share to workspaces		✓		
Manage own code envs		✓		
Manage all code envs		✓		
Manage own cluster		✓		
Manage all clusters		✓		
Manage own Code Studio template		✓		
Manage all Code Studio templates		✓		
Develop plugins		✓		
Edit lib folders		✓		
Create user connections		✓		
Write unisolated code		✓		
Create published API services		✓		
Create published projects		✓		
May write in root project folder		✓		
May create active web content		✓		
Manage Feature Store		✓		
Create Data Collections		✓	✓	✓
Publish to Data Collections		✓	✓	✓

Project authorization

CAR

RPC

WPC

PW

PDC

RD

WD

MAO

MSO

RS

EA

EP

A

RPC

RD

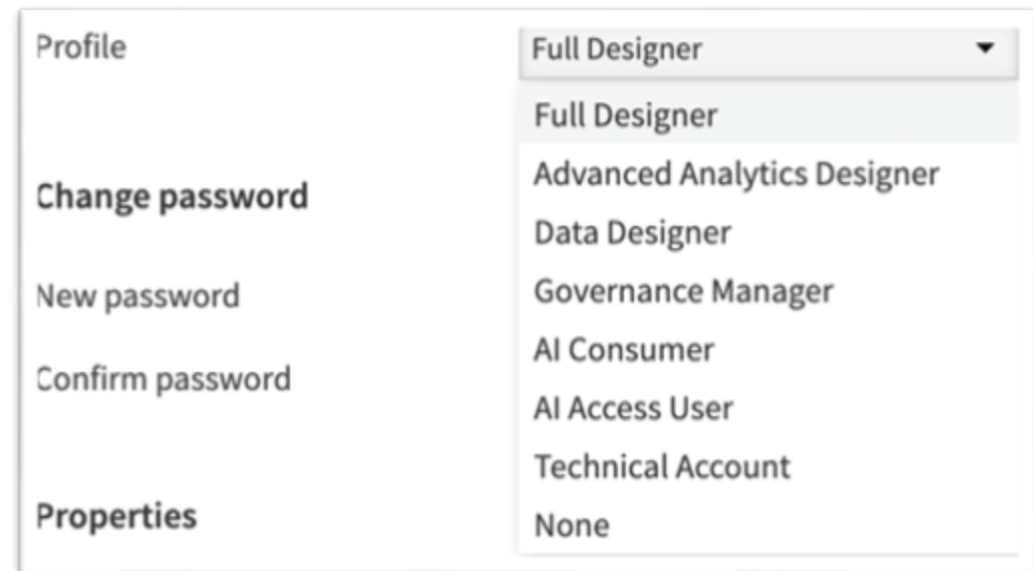
EA

EP

RPC	Read project content	WPC	Write project content	PW	Publish to workspaces	PDC	Publish to Data Collections
RD	Read dashboards	WD	Write dashboards	MAO	Manage authorized objects	MSO	Manage shared objects
RS	Run scenarios	EA	Execute app	EP	Edit permissions	A	Admin

Permission Model > User Profiles

- 모든 DSS user는 하나의 User profile을 가짐
 - **Full-Designer**: 설계·코드·ML·배포까지 **모든 기능 풀액세스**(프로젝트 생성/관리 포함)
 - 통상 Data Scientist
 - Advanced Analytic Designer: 모든 Visual 레시피 사용, Visualization 및 Dataset 생성 가능
 - Governance Manager: Dataiku Govern에서 승인/서명·레지스트리 등 AI 거버넌스 운영
 - **AI Consumer**: 대시보드/웹앱/에이전트허브를 소비(조회·사용)하는 앤드유저 프로파일
 - Technical Account: 자동화·통합 실행용
- 라이선싱을 위한 개념이며 보안을 위한 것이 아님



Permission Model > Global Group Permissions

Edit group "AI_SECTION"

Description

Type

Local

Global permissions

Administrator

☐

Grants all permissions on all scopes

Data Collections

Create Data Collections

☒

Can these users create new Data Collections?

Publish to Data Collections

☒

Can these users publish datasets to Data Collections?

Workspaces

Create workspaces

☐

Can these users create their own workspaces?

Publish on workspaces

☒

Can these users share project objects to workspaces?

Feature Store

Manage Feature Store

☐

Can these users turn datasets into Feature Groups?

Projects creation

Create projects

☒

Can these users create their own projects (blank project, duplication, import)?

Create projects using macros

☒

Can these users create projects using a predefined macro?

Create projects using templates

☒

Can these users create projects using predefined templates, samples, tutorials?

Write in root project folder

☒

Can these users edit root project folder metadata and create projects and subfolders inside it?

Code execution

Write unisolated code

☐

Can these users run local code (Python, R) which runs as DSS service user? Hostile users with this permission may circumvent the permissions system.

Create active Web content

☒

Can these users author Web content that is able to execute Javascript when viewed by other users? This includes webapps, Jupyter notebooks and RMark

Code envs & Dynamic clusters & Code Studio templates

Manage all code envs

☐

Can these users manage all code envs?

Create code envs

☒

Can these users create new code envs?

Manage all clusters

☐

Can these users manage all clusters?

Create clusters

☐

Can these users create new clusters?

Manage all Code Studio templates

☐

Can these users manage all Code Studio templates?

Create Code Studio templates

☐

Can these users create new Code Studio templates?

Advanced permissions

Develop plugins

☒

Can these users create and edit development plugins? Hostile users with this permission may circumvent the permissions system.

Edit lib folders

☐

Can these users edit the global Python & R libraries and the static web resources in the DSS instance? Hostile users with this permission may circumvent t

Create personal connections

☒

Can these users create personal connections? Does not include FS connections

View indexed Hive connections

☐

Can these users view indexed Hive connections in the catalog?

Manage user-defined meanings

☐

Can these users create/delete global UDMs?

Create published API services

☐

Can these users create a published API service?

Create published projects

☐

Can these users create a published project?

Impersonation in webapps

When a webapp runs as a given user U1, and a user U2 connects to it, the webapp can perform API calls to the DSS API on behalf of U2. This requires U1 to

Permission Model > Project 별 Group Permissions

Project owner: pnd22

Project visibility: Inherit global settings (Private)

Project access requests: Inherit global settings (Enabled)

User permissions

Add users to project: By group

Groups: Education, Education2, administrators, readers

Name	Admin	Edit permissions	Publish to Data collections	Publish on workspaces	Export datasets	Read dashboards	Write dashboards	Run scenarios	Manage authorized objects	Manage shared objects
data_team	☐	☒	☒	☒	☒	☒	☒	☒	☒	☐

- 각 Project에 대해서 어떤 Group이 Access(R/W)할 수 있는지 또한 어떤 권한을 부여할 지 설정 할 수 있음

Permission Model > Project 별 User Permissions

The screenshot shows the 'Project security' settings for a project named 'car'. The 'Project security' tab is highlighted in the top navigation bar. On the left sidebar, 'Permissions' is selected. The main content area shows settings for 'Project owner' (pnd22), 'Project visibility' (Inherit global settings (Private)), and 'Project access requests' (Inherit global settings (Enabled)). Below these, the 'User permissions' section is visible. It includes a dropdown menu for 'Add users to project' set to 'By user', a search box with '- Select users -', and a '+ GRANT ACCESS' button. A table lists the permissions for two users: 'data_team' and 'pnd00'. The table columns include Name, Admin, Edit permissions, Read project content, Write project content, Publish to Data Collections, Publish on workspaces, Export datasets, Read dashboards, Write dashboards, Run scenarios, Manage authorized objects, and Manage shared objects. The 'data_team' user has permissions for Admin, Edit permissions, Read project content, Write project content, Publish to Data Collections, Publish on workspaces, Export datasets, Read dashboards, Write dashboards, Run scenarios, and Manage authorized objects. The 'pnd00' user has permissions for Read project content, Write project content, Publish to Data Collections, Read dashboards, Write dashboards, Run scenarios, and Manage authorized objects.

Name	Admin	Edit permissions	Read project content	Write project content	Publish to Data Collections	Publish on workspaces	Export datasets	Read dashboards	Write dashboards	Run scenarios	Manage authorized objects	Manage shared objects
data_team	☐	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☐
pnd00	☐	☐	☒	☐	☒	☐	☐	☒	☐	☐	☒	☐

- 각 Project에 대해서 Group 전체가 아닌 개별 User 레벨로도 Access(R/W) 여부와 또한 어떤 권한을 부여할 지 설정 할 수 있음

Permission Model > Data Connections

- Data Connection은 접근할 수 있는 Group 제한을 해야 함
- Admin은 Connection들에 대해서 Group 별로 사용가능 여부와 조회가능 여부를 설정할 수 있음
- Connection은 read only로 생성될 수 있음
- 어떤 Connection들은 개별 User별 인증 설정을 지원 함

USAGE PARAMS

Allow write	☒	Can DSS write to the datasets from this connection?
Allow managed datasets	☒	Can new managed datasets be created on this connection?
Allow managed folders	☒	Can new managed folders be created on this connection?
Max nb. of activities	0	Limit the number of concurrent activities involving

SECURITY SETTINGS

See [the documentation](#) for more information.

Freely usable by	☐ Every analyst ☒ Selected groups
Who can create new datasets in this connection, and more generally "browse" this connection.	
↔ usable by groups	administrators, data_team, reade
Details readable by	☒ Nobody ☐ Every analyst ☐ Selected groups
Who can access the details of the connection (including credentials, if connection has some). Note t from Spark	

CREDENTIALS

See [the documentation](#) for more information.

Credentials mode	Per user
------------------	---------------------------------------

- SSL 인증서 설정을 통해서 HTTPS로 전환 가능

- 설정 파일: <DATA_DIR>/install.ini
- 설정 방법:

```
[server]
```

```
port = 10010
```

```
ssl = true
```

```
ssl_certificate = <인증서 파일 PATH>
```

```
ssl_certificate_key = <인증키 파일 PATH>
```

```
ssl_ciphers = recommended
```

- 접속 URL: https://DSS_HOST:DSS_PORT

Lab. 사용자 생성

The screenshot shows the DSS settings interface. The top navigation bar includes 'Workspaces', 'Local Deployer', 'License', 'Connections', 'Security' (highlighted with a red arrow), 'Clusters', 'Code Studios', 'Code Envs', 'Monitoring', 'Maintenance', and 'Settings'. The left sidebar shows 'DSS settings' with a sub-menu 'Users' (highlighted with a red arrow). The 'New user' form is displayed, with a red box highlighting the 'General' section. The 'General' section contains the following fields:

General	
Login	gildong
Type	Local
Display name	홍길동
Email	gildong@abc.com
Groups	data_team
Profile	Full Designer
Password	
Password	
Confirm password	

The 'SAVE' button is located at the top right of the form, highlighted with a red arrow.

Lab. Group 생성

Dataiku Search DSS...

License Connections Security Clusters Code Studios Code Envs Monitoring Maintenance Settings

Users
Groups
Global API keys
Personal API keys
Authorization Matrix
Audit trail

New group

Group name* AI_SECTION

Description AI 전문가 그룹

Type Local

SAVE

Global permissions

Administrator ☐ Grants all permissions on all scopes

Data Collections

Create Data Collections ☒ Can these users create new Data Collections?

Publish to Data Collections ☒ Can these users publish datasets to Data Collections?

Code envs & Dynamic clusters & Code Studio templates

Manage all code envs ☐ Can these users manage all code envs?

Create code envs ☒ Can these users create new code envs?

Manage all clusters ☐ Can these users manage all clusters?

Create clusters ☐ Can these users create new clusters?

Manage all Code Studio templates ☐ Can these users manage all Code Studio templates?

Create Code Studio templates ☐ Can these users create new Code Studio templates?

Lab. Group에 사용자 추가

Users

- Groups
- Global API keys
- Personal API keys
- Authorization Matrix
- Audit trail

Edit user "gildong"

General

Type: Local

Display name: 홍길동

Email: gildong@abc.com

Groups: data_team

Profile: Filter...

Change password

New password:

Confirm password:

Groups dropdown:

- AI_SECTION
- Education
- Education2
- administrators
- data_team ✓
- readers

SAVED!

Lab. Group에 사용자 추가

Users

- Groups
- Global API keys
- Personal API keys
- Authorization Matrix
- Audit trail

Edit user "gildong"

General

Type: Local

Display name: 홍길동

Email: gildong@abc.com

Groups: data_team

Profile: Filter...

Change password

New password:

Confirm password:

Groups List:

- AI_SECTION
- Education
- Education2
- administrators
- data_team ✓
- readers

SAVED!

Lab. Per User Connection

DSS settings

PostgreSQL connection: PerUserConn

Basic params

Host: 175.212.202.76
Database: dataiku
Port: 15432

Advanced JDBC properties

+ ADD PROPERTY

Use custom JDBC URL: ☐ Used to customize JDBC connection URL

Advanced params

Enable PostGIS support: ☐ PostGIS extension should be installed in a PostgreSQL database

Driver to use: Dataiku-managed

Send driver jars to Spark: ☐ Experimental: Add driver jars to spark-submit subprocesses

Fetch size: Empty = default

Schema search path:

This search path will be set on all sessions. Useful for SQL queries and notebooks

Post-connect commands: These statements are run after the connection is created

Job post-connect commands: When used in a job context, these additional statements are run after the connection is created

Truncate to clear data: ☐ When the columns did not change, should DSS truncate tables instead of dropping them?

Indexing

These settings apply if you choose to index this connection to make it searchable in the DSS data catalog

Index indices: ☐ Fetch and index the indices in the database? May strongly increase indexing time

Index foreign keys: ☐ Fetch and index foreign key relationships in the database? May strongly increase indexing time

Index system tables: ☐ Fetch and index system tables?

Credentials

See the documentation for more information.

Credentials mode: Per user

Custom provider (advanced): Class name of a custom basic credential provider

Naming rules for new datasets

These settings define how managed SQL datasets are mapped to database tables. These settings are applied when creating a new managed SQL dataset. You can always modify these afterwards in the SQL dataset settings. They are also used to create temporary tables if the connection is used for bundled data. See the documentation for more information.

Schema:

Managed datasets will be created in this schema. You may reference variables here, they will be expanded dynamically

Allow schema override: ☐ If this is enabled, an option to choose a schema will be presented to users each time they create a managed dataset on this connection

Table prefix: \${projectKey}_

Table name for new managed datasets will be prefixed by this. You may reference variables here, they will be expanded dynamically

Table suffix:

Table name for new managed datasets will be suffixed by this. You may reference variables here, they will be expanded dynamically

Date settings

Default assumed timezone: Local time zone

Default "assumed time zone" selection when adding a dataset through this connection. This will be the default timezone assigned to timezone-less data when read in as a DSS datetime with tz.

Custom properties

Usage params

Allow write: ☒ Can DSS write to the datasets from this connection?

Allow managed datasets: ☒ Can new managed datasets be created on this connection?

Max nb. of activities: 0

Limit the number of concurrent activities involving this connection.

Security settings

See the documentation for more information.

Freely usable by: ☒ Selected groups

Who can create new datasets on this connection, and more generally "browse" this connection?

AI_SECTION

usable by groups: AI_SECTION

Details readable by: ☒ Nobody

Who can access the details of the connection (including credentials, if connection has some).

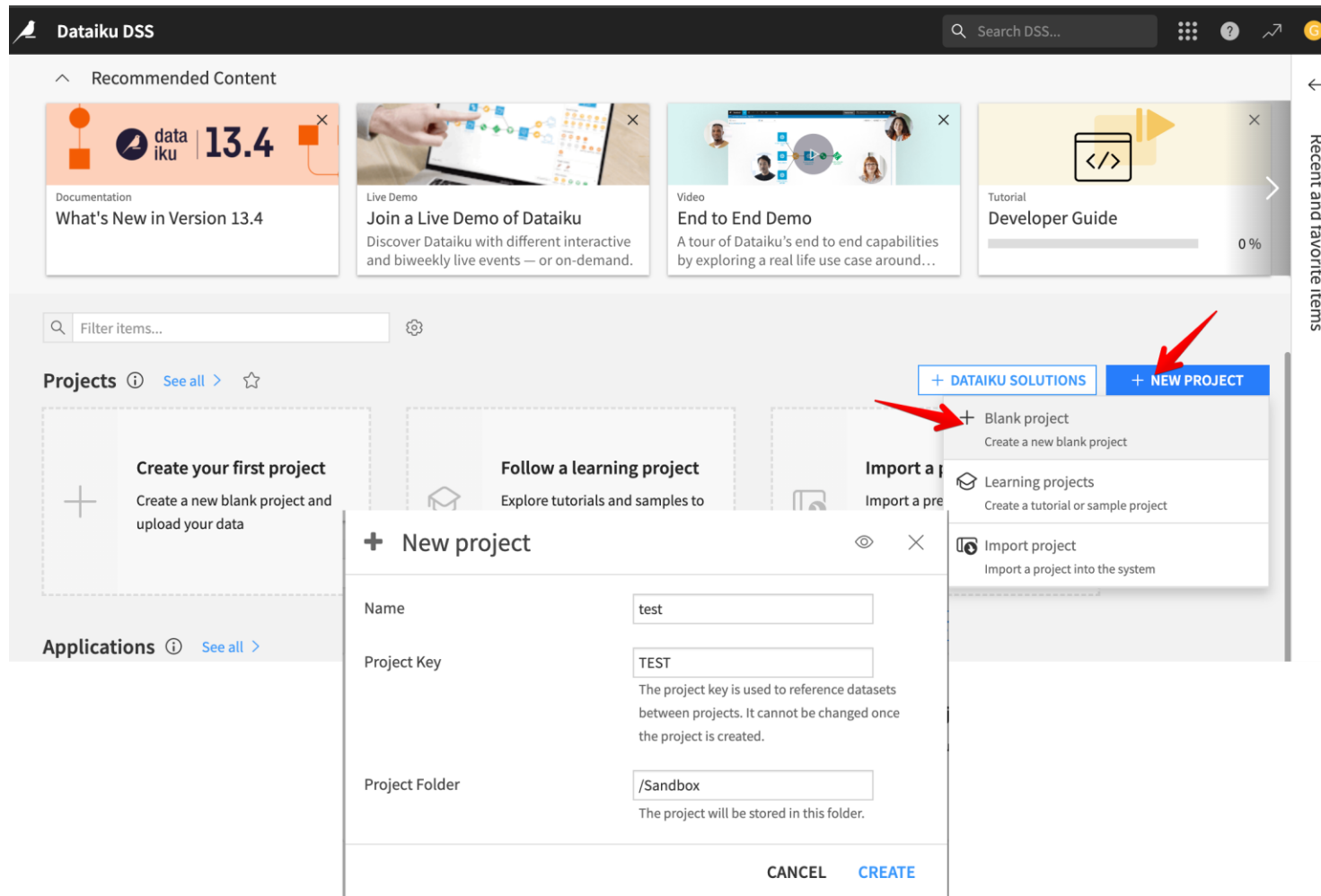
❖ 새로운 Postgresql DB 연결 생성

- Name: PerUserConn
- Host, Database, Port는 이전 Lab과 동일
- 화면 아래로 scroll down
- 우측 Security settings > Freely used by : Selected groups 선택
- usable by groups에선 생성한 group 선택



**다른 종류의 Browser에서
새로 생성한 User 로 로그인 후 진행**

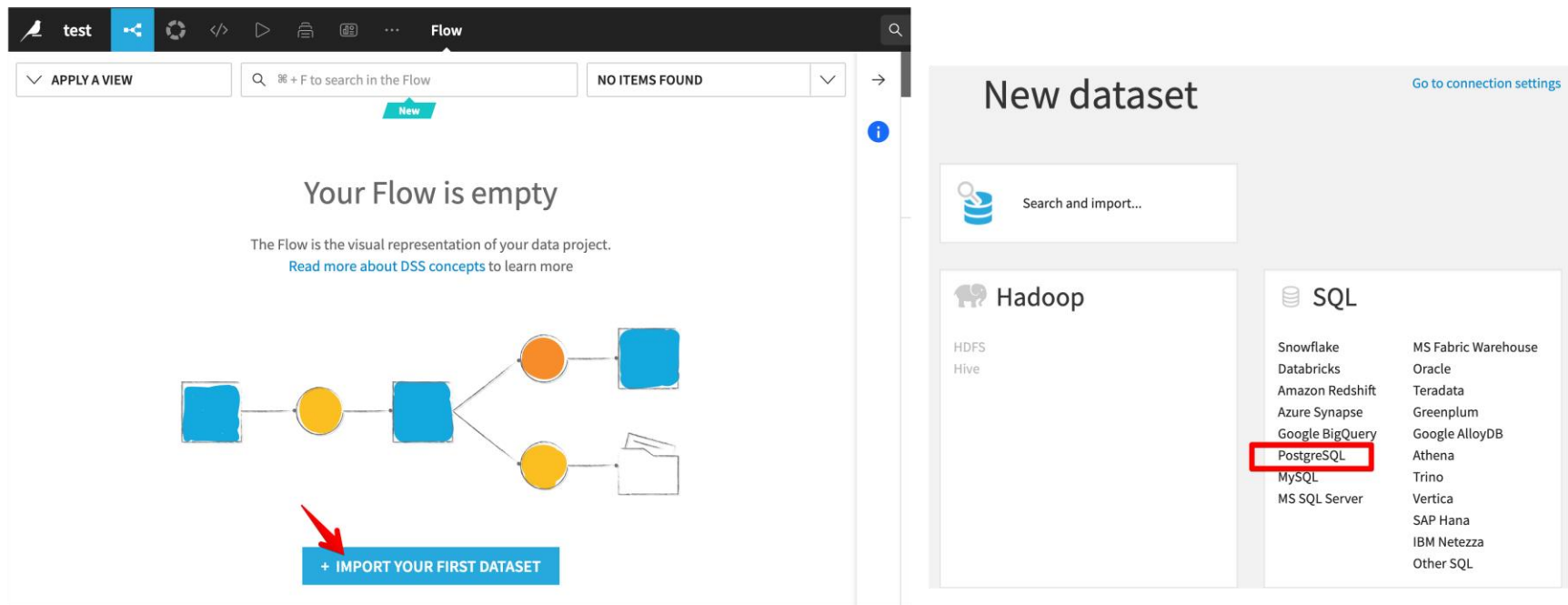
Lab. Per User Connection



❖ Test project 생성

- 사용자 계정(gildong) 으로 DSS 접속
- NEW Project
- Blank Project
- Project Name: test

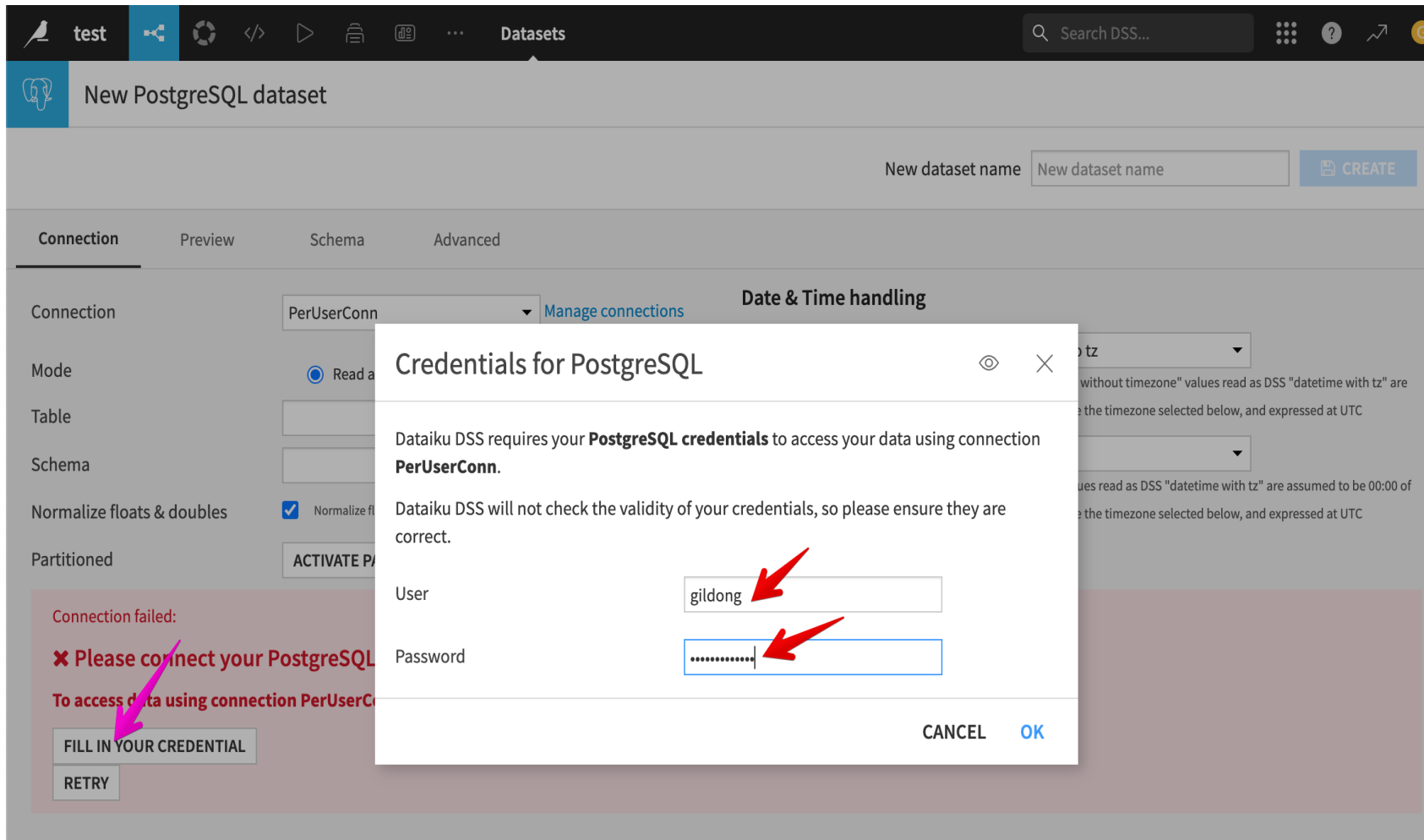
Lab. Per User Connection



❖ Import Dataset

- SQL > PostgreSQL 선택

Lab. Per User Connection



❖ 사용자 인증 정보 입력

- FILL IN YOUR CREDENTIAL 클릭
- User: gildong
- Password: pndsolution1

Lab. Per User Connection

Mode: ☒ Read a database table ☐ SQL query

Table:

Schema:

Normalize floats & doubles: ☐

Partitioned: ☐

Connection failed:
❌ Please connect your PostgreSQL account to continue
To access data using connection PerUserConn, you need to connect
✅ credential saved!
RETRY

education
information_schema
pg_catalog
public

Project home PostgreSQL dataset

New dataset name: test_scores
CREATE

Connection: PerUserConn
Manage connections

Mode: ☒ Read a database table ☐ SQL query

Table: test_scores (public)
ENTER CUSTOM

Normalize floats & doubles: ☒ Normalize floating point values (force '42' to '42.0')

Partitioned: ☐ ACTIVATE PARTITIONING

Connection and table OK
TEST AGAIN

Date & Time handling

Read SQL "datetime without timezone" as: Datetime no tz
SQL "datetime without timezone" values read as DSS "datetime with tz" are assumed in the the timezone selected below, and expressed at UTC

Read SQL "date" as: Date only
SQL "date" values read as DSS "datetime with tz" are assumed to be 00:00 of that date in the the timezone selected below, and expressed at UTC

Preview

id	student_name	score
1	Alice	85
2	Bob	92
3	Charlie	78
4	David	88
5	Eva	95
6	Frank	73

❖ 추가 DB 정보

- Schema ← 자동 조회 버튼 활용
- Table ← Get Table List 버튼 활용
- TEST TABLE 버튼을 통해 Data 확인
- 화면 우측 상단 CREATE 버튼 클릭

Lab. Per User Connection

test

test_scores

Explore

Charts

Statist

Whole data

10 rows

id	student_n...	score
int	string	int
Integer	Text	Integer
1	Alice	85
2	Bob	92
3	Charlie	78

G

홍길동

@gildong

0 unread discussions

You have no notification

Dataiku DSS

Search DSS...

?

G

홍길동

My Settings

Exports

Stars and watches

API keys

Credentials

My files

My Account

LOGOUT

G

Personal credentials

Connections

Connect your account to your external data through the Dataiku platform. You can either insert ID/PASSWORD right away, or connect to the 3rd party right away depending on the setup by admin.
If you have questions, see the documentation or contact your administrator.

Connection type	Connection name	Type	Status	Actions	Last update
PostgreSQL	PerUserConn	Password	Filled (gildong)	UPDATE REMOVE	Today at 14:37

❖ 사용자 Profile 확인

- 화면 우측 상단 User Profile 아이콘 클릭
- 기어 아이콘 클릭
- User profile 화면에서 "Credentials" 탭
- Connections 에서 추가된 DB Credential 확인



5. 시스템 운영 관리



5. 시스템 운영 관리 - Logs

- 로그의 종류
 - Main DSS process logs
 - Job logs
 - Scenario logs
 - Analysis logs
 - Audit logs

DSS Logs > Main DSS process logs

Log file	로그 내용
backend.log	DSS backend process의 로그 모든 사용자의 DSS UI와 API를 통한 모든 Operation들을 로깅 함 Visual 분석의 학습세션 내용도 여기에 로깅 됨
hproxy.log	Hadoop 연결 관련 내용
nginx.log	nginx HTTP Server이 일반 로그, 사용자 Trace 정보는 포함하지 않음
nginx/access.log	nginx HTTP 서버에 대한 Access 로그
ipython.log	R, Python 등의 노트북에서 사용되는 Jupyter Server 로그
supervisor.log	프로세스 제어 로그, 모든 관리 프로세스들에 대한 Start/Stop을 로깅함
frontend.log	사용자 브라우저의 Javascript 동작 로그

DSS Logs > Main DSS process logs

- 로그 파일 위치

- <DATA_DIR>/run

- UI:

- Administration>Maintenance>Log files

```
dataiku@pndiku: /data/dataiku/design/run (ssh)  %1  pndiku@pndiku-desktop: ~ (ssh)
drwxr-xr-x  2 dataiku dataiku   4096 Oct 26 00:57 audit
-rw-r--r--  1 dataiku dataiku 1714540 Oct 26 12:43 backend.log
-rw-r--r--  1 dataiku dataiku    496 Oct 26 07:34 dbservers.json
-rw-r--r--  1 dataiku dataiku 153066 Oct 26 12:40 frontend.log.0
-rw-r--r--  1 dataiku dataiku     0 Oct 26 00:59 frontend.log.0.lck
-rw-r--r--  1 dataiku dataiku   4128 Oct 26 00:57 install.log
-rw-r--r--  1 dataiku dataiku 304215 Oct 26 12:43 ipython.log
drwxr-xr-x  2 dataiku dataiku   4096 Oct 26 05:33 nginx
-rw-r--r--  1 dataiku dataiku    333 Oct 26 07:34 nginx.log
-rw-r--r--  1 dataiku dataiku    64 Oct 26 00:57 shared-secret.txt
-rw-r--r--  1 dataiku dataiku  3711 Oct 26 07:34 supervisord.log
-rw-r--r--  1 dataiku dataiku     5 Oct 26 07:34 supervisord.pid
srwx-----  1 dataiku dataiku     0 Oct 26 07:34 svd.sock
drwxr-xr-x  2 dataiku dataiku   4096 Oct 26 00:59 unified-monitoring
-rw-r--r--  1 dataiku dataiku    151 Oct 26 08:04 user-last-activity.json
-rw-r--r--  1 dataiku dataiku    361 Oct 26 01:05 user-sessions.json
dataiku@pndiku: /data/dataiku/design/run$
```

 DSS settings

License

Connections

Security

Clusters

Code Studios

Code Envs

Monitoring

Maintenance

Settings

System info

Instance diagnosis

Instance sanity checks

Scheduled tasks

Log files

DSS logs

If you want to report an issue to Dataiku Support, please do not send these logs. Instead, use "Instance diagnosis"

Main log

DOWNLOAD BACKEND.LOG

All log files

DOWNLOAD ALL LOG FILES (.ZIP)

Log file ^	Last modified	Size
audit/audit.log	just now	205.01 KB
backend.log	just now	1.63 MB
frontend.log.0	just now	149.37 KB
install.log	11 hours ago	4.03 KB
ipython.log	just now	295.62 KB
nginx.log	5 hours ago	0.33 KB
nginx/access-2025-10-26-05-33-50-704.log.gz	7 hours ago	1.94 KB
nginx/access.log	just now	110.7 KB
supervisord.log	5 hours ago	3.62 KB

DSS Logs > Main DSS process logs

- 로깅 설정 : <DATA_DIR>install.ini
 - 기본 설정: 최대 50MB , 최대 10개 파일
- 변경 절차

```
# Stop DSS
DATADIR/bin/dss stop
# Edit installation options
vi DATADIR/install.ini
# Regenerate DSS configuration according to the new settings
DATADIR/bin/dssadmin regenerate-config
# Restart DSS
DATADIR/bin/dss start
```



```
[logs]
# Maximum file size, default 50MB.
# Suffix multipliers "KB", "MB" and "GB" can be used in this value.
# Define as 0 to disable automatic log file rotation.
logfiles_maxbytes = SIZE
# Number of retained files, default 10.
logfiles_backups = NUMBER_OF_FILES
```

Job Logs

- Recipe 실행 시 마다 log file이 생성 됨
 - Project>상단 삼각형 Play 버튼 클릭 , 단축키: "gj"

The screenshot displays the Dataiku DSS interface. On the left, a sidebar shows a list of jobs with filters for 'Newest', 'Any status', and 'All users'. The main area shows a detailed view of a specific job: 'Build car_data_1_Sheet1_joined_prepared2 (NP)' executed on Friday, 26 September 10:57 with a duration of 3s. Below the job title, a workflow diagram is visible, showing data sources '1_Sheet1' and '2_Sheet1' feeding into a 'Join' node, followed by a 'Prepare' node, and then a 'Split' node leading to two output folders labeled '2024' and '2025'. The '2025' folder is further processed by a 'Predict 가격 (regression)' node. At the bottom, a '1 Activity' section shows a single activity: 'compute_car_data_1_Sheet1_joined_prepared2' with a green status icon.

```
dataiku@ndiku:/data/dataiku/design/jobs/CAR$ ls -al
total 12
drwxr-xr-x 3 dataiku dataiku 4096 Oct 26 13:02 .
drwxr-xr-x 3 dataiku dataiku 4096 Oct 26 13:02 ..
drwxr-xr-x 4 dataiku dataiku 4096 Oct 26 13:02 Build_from_compute_car_data_1_Sheet1_joined_2025-10-26T13-02-42.408
dataiku@ndiku:/data/dataiku/design/jobs/CAR$
```

- UI는 최근 100개 job log만 보여 줌
- 전체 로그 경로: <DATA_DIR>/jobs/PROJECT_KEY/

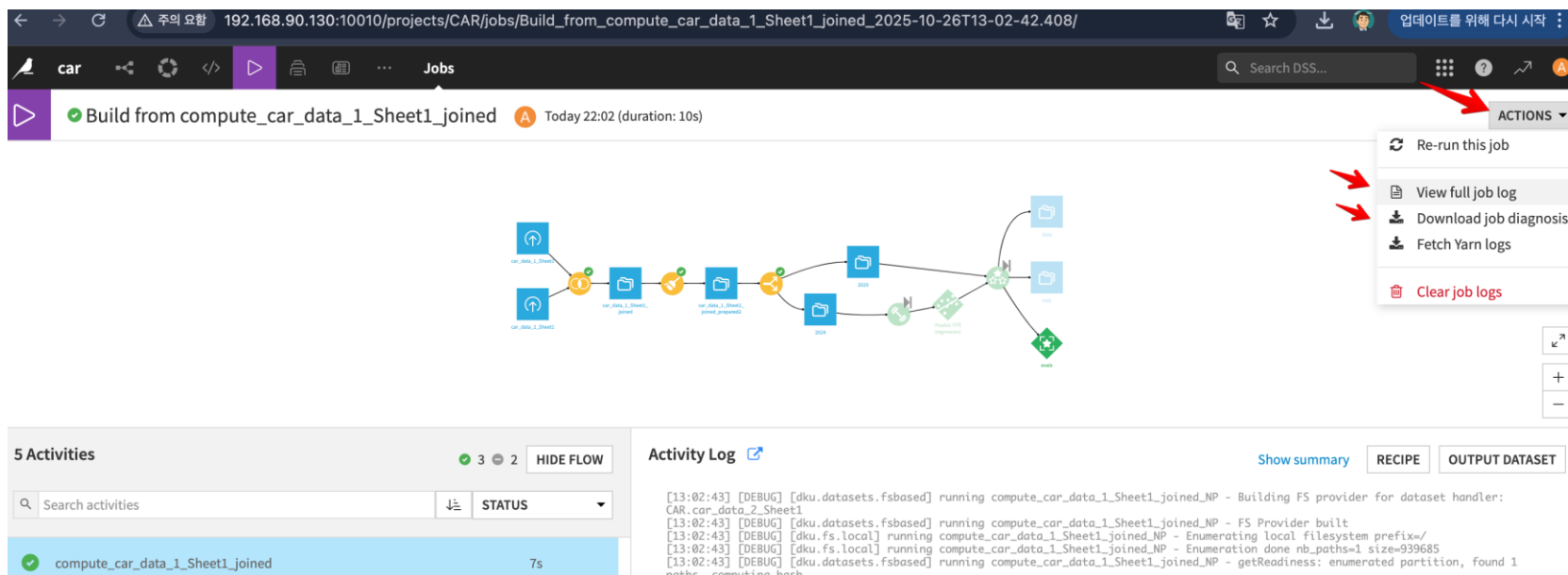
Lab. 프로젝트 실습

❖ Task

- CAR.zip 이용 프로젝트 Import
- Build project

Job Logs

- “Actions” 버튼 클릭 후 Full log 보기 및 download 가능

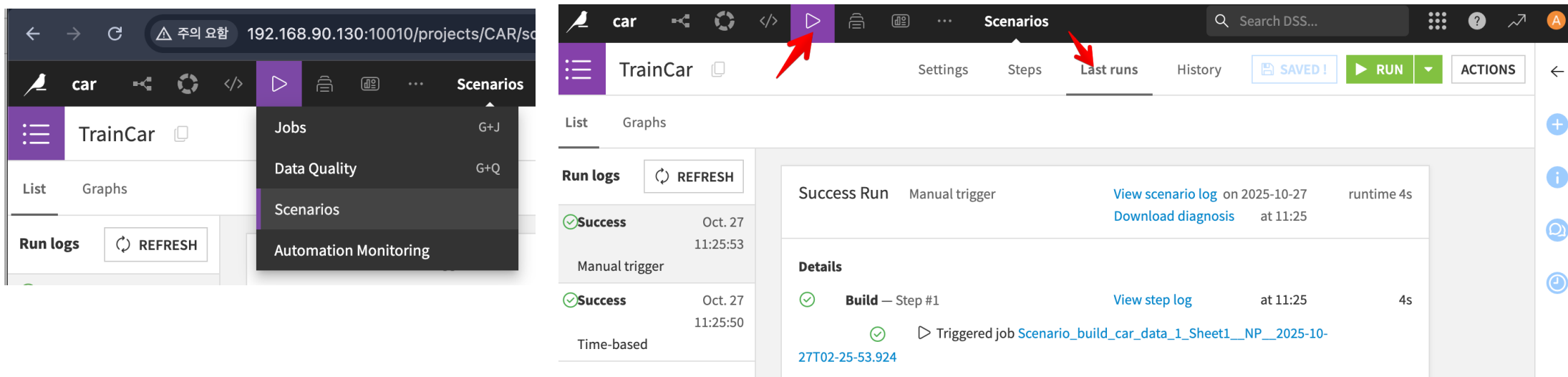


- Dataiku Support 진행 시, Job diagnosis를 down 받아 전달하면 빠른 이슈 해결에 도움
- DATA_DIR/job/PROJECT_KEY log 파일은 명시적으로 purge 해야 함
- Macros 사용이 도움이 됨 ← 뒤에서 별도 설명함

Scenario Logs

- 시나리오 실행 시 로그

- 매 실행 시마다 프로젝트설정/플로우/코드 스냅샷 생성 됨
- 실행 결과에 대한 로그 및 진단 정보 보관



- UI 로그: Play 버튼 > Scenarios > 시나리오 선택 > Last runs
- 서버 로그 경로: <DATA_DIR>/scenarios/PROJECT_KEY/SCENARIO_ID/SCENARIO_RUN_ID

Audit Trail Logs

- 감사 추적(Audit Trail) 로그
 - 사용자 수행 모든 액션을 기록: 사용자ID,타임스탬프, IP, 인증방식 등 메타데이터 포함
 - UI 경로: Administration > Security > Audit trail
 - UI가 표시하는 라이브 뷰는 최근 1,000건만 보관
- 실제 감사 운영 옵션
 - 로그파일 사용: DATA_DIR/run/audit
 - 또는 외부로그/보안시스템(SIEM 등) 으로 연동

DSS settings

License Connections **Security** Clusters Code Studios Code Envs Monitoring Maintenance Settings

Users
Groups
Global API keys
Personal API keys
Authorization Matrix
Audit trail

Audit trail ☐ Include all calls

Warning: this live view only shows the last elements added in the audit log since the last DSS restart.
For complete audit data, see documentation.

Timestamp	Type	User	IP	Details
2025/10/27-11:45:03		admin	192.168.90.1	requestedSize: 24x24, type: USER, id: gildong, hash: 141069, time: 1, xForwardedFor: 192.168.90.1
2025/10/27-11:45:03		admin	192.168.90.1	time: 1, xForwardedFor: 192.168.90.1
2025/10/27-11:45:03		admin	192.168.90.1	lastActivity: true, time: 1, xForwardedFor: 192.168.90.1
2025/10/27-11:45:03	✓ security-admin-user-get-syncable-source-types	admin	192.168.90.1	xForwardedFor: 192.168.90.1
2025/10/27-11:45:03	✓ security-admin-users-list	admin	192.168.90.1	xForwardedFor: 192.168.90.1

```
dataiku@pndiku: /data/d
dataiku@pndiku: /data/dataiku/design/run/audit (ssh)
dataiku@pndiku:/data/dataiku/design$ cd run/audit
dataiku@pndiku:/data/dataiku/design/run/audit$ ls -al
total 492
drwxr-xr-x 2 dataiku dataiku 4096 Oct 26 00:57 .
drwxr-xr-x 5 dataiku dataiku 4096 Oct 27 02:21 ..
-rw-r--r-- 1 dataiku dataiku 489733 Oct 27 02:45 audit.log
```

- 로그 설정 경로

- <Install_DIR>/resources/logging/dku-log4j.properties

- **주의** 권고

- Dataiku Support 팀의 확인 없이 로그 레벨 변경은 권장 하지 않음.

Customizing log levels

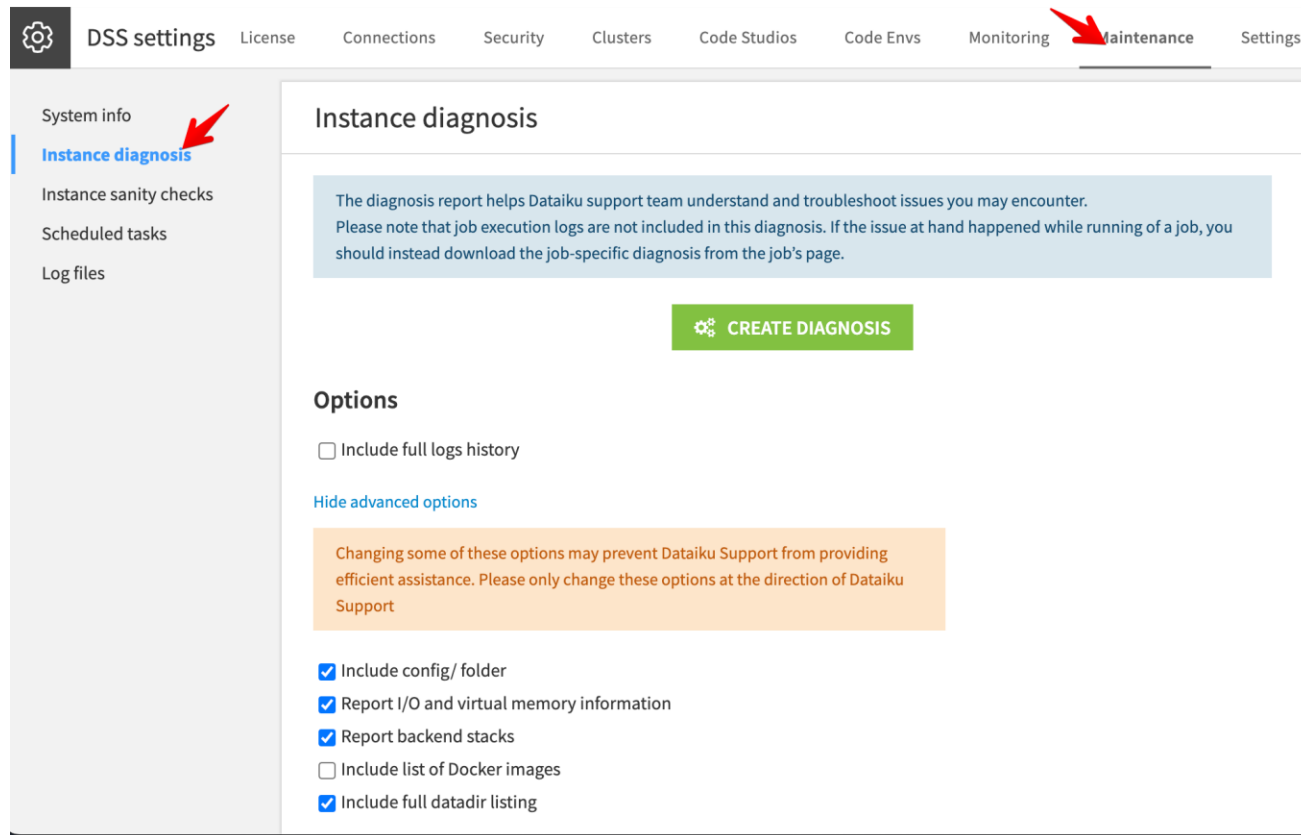
Warning

We strongly recommend that you **do not** customize log levels without input from Dataiku Support.

Modifying or lowering the logging levels can potentially reduce the ability of Dataiku Support to investigate / debug certain issues and result in an overall degraded support experience. In these situations, you will likely be asked to revert these logging level changes and reproduce the issue when further investigation is required.

Diagnostic Tool

- DSS Support 를 통해 이슈 해결 시 활용
 - Dataiku 설정을 포함한 이슈 해결에 꼭 필요한 정보를 하나의 파일로 생성함
 - UI 경로: Administration > Maintenance > Instance diagnosis





5. 시스템 운영 관리 - Troubleshooting

- UI Down 시

- DSS process 상태 점검
- tail 명령어 등으로 backend.log 확인, 로그파일 위치:<DATA_DIR>/run/
- *Exceptions [ERROR] 검색과 stacktraces 확인
- 만약 dataset 관련이라면, network 통신 가능 여부 확인

- UI 접근 가능 시

- 웹 UI를 통한 backend.log 점검 (Administration>maintenance>Log files>backend.log)
- *Exceptions [ERROR] 검색과 stacktraces 확인
- 다른 Project에서 동일한 조작
- 만약 dataset 관련이라면, connection 테스트 수행

Troubleshooting > Job Issue

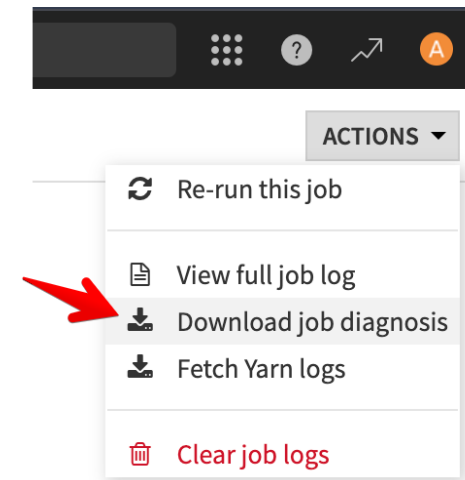
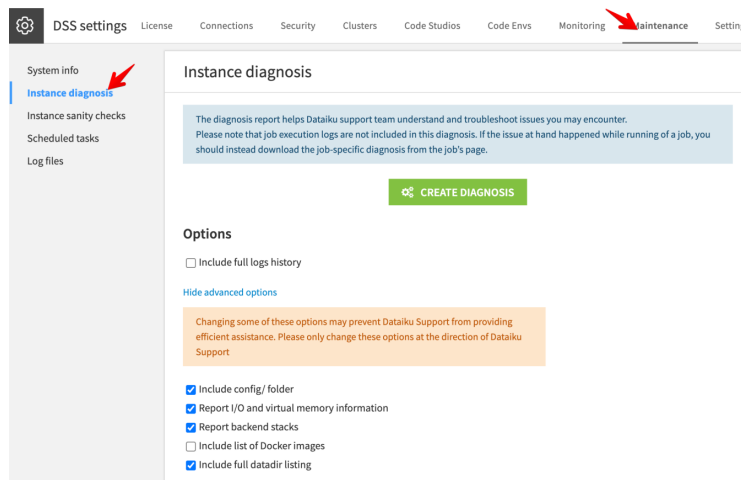
● Job 이슈 대응

- Exception Stracktrace 확인 - 특히, 'caused by section' 우선 점검
- 모든 외부 Connection 정상 여부 점검
- 가급적 "notebook" 으로 Step-by-Step 점검
- backend.log 에서 실행 한 "command" 체크

The screenshot displays the Dataiku job interface. On the left, the '2 Activities' panel shows a list of activities: 'split_car_data_1_Sheet1' with a red 'X' icon and '0s' duration, and 'compute_car_data_1...' with a green checkmark and '0s' duration. The 'split_car_data_1_Sheet1' activity is selected. On the right, the 'Activity Log' panel shows a red error message: 'Oops: an unexpected error occurred'. Below the message, it states 'Splitting mode is not implemented' and 'Please see our options for getting help'. The HTTP code is 'type: com.dataiku.dip.utils.NotImplementedException'. Below the error message, the log shows the following information: [01:52:35] [INFO] [dku] running split_car_data_1_Sheet1_prepared_NP - -----, [01:52:35] [INFO] [dku] running split_car_data_1_Sheet1_prepared_NP - DSS startup: jek version:14.0.0, [01:52:35] [INFO] [dku] running split_car_data_1_Sheet1_prepared_NP - DSS home: /home/dataiku/dss, [01:52:35] [INFO] [dku] running split_car_data_1_Sheet1_prepared_NP - OS: Linux 6.8.0-65-generic amd64 - Java: Red Hat, Inc. 17.0.15, [01:52:35] [INFO] [dku.flow.jobrunner] running split_car_data_1_Sheet1_prepared_NP - Allocated a slot for this activity! Below the log, the 'Activity Log' panel shows a detailed stack trace for the error: com.dataiku.dip.utils.NotImplementedException: Splitting mode is not implemented at com.dataiku.dip.dataflow.exec.split.SplitRecipeStreamExecutor.run(SplitRecipeStreamExecutor.java:168) at com.dataiku.dip.dataflow.exec.split.SplitRecipeRunner.run(SplitRecipeRunner.java:128) at com.dataiku.dip.dataflow.jobrunner.ActivityRunner\$FlowRunnableThread.run(ActivityRunner.java:428) [01:52:36] [INFO] [dku.flow.activity] running split_car_data_1_Sheet1_prepared_NP - activity is finished [01:52:36] [ERROR] [dku.flow.activity] running split_car_data_1_Sheet1_prepared_NP - Activity failed com.dataiku.dip.utils.NotImplementedException: Splitting mode is not implemented at com.dataiku.dip.dataflow.exec.split.SplitRecipeStreamExecutor.run(SplitRecipeStreamExecutor.java:168) at com.dataiku.dip.dataflow.exec.split.SplitRecipeRunner.run(SplitRecipeRunner.java:128)

- Support 전달 자료

- DSS diagnostic
- Job diagnostic
- System 정보



- UI down 시

- command 를 통해 diagnosis 자료 생성 가능
- <DATA_DIR>/bin/dssadmin
run-diagnosis -cfls /tmp/dataiku_diag.zip

```
dataiku@pndiku: /data/dataiku/
dataiku@pndiku: /data/dataiku/design/bin (ssh)
Usage : dssadmin run-diagnosis [OPTIONS] OUTPUT_FILE
Options:
  -c : include config dir
  -s : include backend stacks
  -d : include list of Docker images
  -f : include full logs
  -l : include full listing of data dir
  -e : include service endpoints files (except data)
dataiku@pndiku: /data/dataiku/design/bin$
```

- Support 요청 방법
 - support portal 을 통해 ticket open :
 - <https://support.dataiku.com>
 - email 전송:
 - support@dataiku.com



5. 시스템 운영 관리 - Disk 관리

- Why

- Dataiku DSS 하위에 여러 디렉토리에 Disk 사용량이 증가 함
- 자동으로 rotation 되지 않는 파일을 Admin 판단으로 삭제 해야 함
- 예를 들어 Job log는 사용자가 과거 일정 시점까지 보관하기 원함

- How to

- cron 작업 등으로 매뉴얼 하게 삭제
- DSS Macro를 Scenario에서 실행

Disk 관리 > 대상 파일 종류

- 관리 대상 파일

- Job 실행 로그: **jobs/**
- Scenario 실행 로그: **scenarios/**
- Analysis Data : **analysis-data/ANALYSIS_ID/MLTASK_ID**
- Saved Models : **saved_models/PROJET_KEY/SAVED_MODEL_ID/version/VERSION_ID**
- Export files : **exports/**
- Temporary files : **tmp/**
- Caches : **caches/**

```
dataiku@pndiku: /data/dataiku/design
dataiku@pndiku: /data/dataiku/design (ssh)
pndiku@pndiku-desktop: ~ (-zsh)

dataiku@pndiku:/data/dataiku/design$ ls -ld analysis-data saved_models exports tmp caches jobs scenarios
drwxr-xr-x 3 dataiku dataiku 4096 Oct 26 12:57 analysis-data
drwxr-xr-x 7 dataiku dataiku 4096 Oct 27 07:14 caches
drwxr-xr-x 5 dataiku dataiku 4096 Oct 26 00:57 exports
drwxr-xr-x 3 dataiku dataiku 4096 Oct 26 13:02 jobs
drwxr-xr-x 3 dataiku dataiku 4096 Oct 26 12:57 saved_models
drwxr-xr-x 3 dataiku dataiku 4096 Oct 27 02:25 scenarios
drwxr-xr-x 9 dataiku dataiku 4096 Oct 26 13:01 tmp
dataiku@pndiku:/data/dataiku/design$
```

- 정의: Task 실행 자동화를 위해 미리 정의된 액션들

- 유지 관리 및 진단(diagnostic) task
- Data 가져오기를 위한 특정 연결 task
- 데이터 또는 DSS에 대한 보고서 생성 task 등

- Macro 실행 옵션

- 수동: Project의 Macro 화면에서 수동 실행
- 자동: Scenario Step을 통해서 자동 실행
- 대시보드에 사용자를 추가하여 대시보드 사용자가 실행

- Macro 종류

- DSS 기본 제공 Macro
- 사용자 개발 Macro

- 어떤 연결이 사용되는지에 대한 감사 보고서를 생성
- 태그 필터로 데이터세트 나열 및 대량 삭제
- 내부 DSS 데이터베이스 지우기
- 이전 DSS 작업 로그 지우기
- 너무 오랫동안 실행 중이거나 유휴 상태인 Jupyter 세션을 종료

DSS Macros > DSS 제공

The image shows two overlapping screenshots of the DSS (Data Science Studio) interface. The top-left screenshot shows the 'Flow' menu with 'Macros' highlighted by a red arrow. The bottom-right screenshot shows the 'Macros' page with a search bar and several macro categories: Audit, builtin macros, and DSS maintenance.

Flow Menu:

- API Designer
- Bundles
- Variables
- Macros**
- Version Control
- Application Designer
- Settings
- Security

Macros Page:

Macros are one-shot actions, aimed at the maintenance of your project. Click on a macro to enter parameters and run it. Macros can also be run from multiple DSS elements. Please see [the documentation](#) for more information.

Search... Categories

Audit

- Connections audit**
List datasets and notebooks of the project, per connection

builtin macros

- Delete datasets by tag**
Delete all datasets matching a tags filter (included or excluded)
- Train paused ML task queues**
Trains all non-running, non-empty ML task queues at the ML task or project level

DSS maintenance

- Backup internal databases**
This macro backs up internal DSS databases to Zip files
- Clear app instances**
Delete old instances of apps
- Clear assets from expired trial users**
Delete assets from trial users whose trials have expired and did not become permanent users at the end of their trials.
- Clear continuous activities logs**
Delete logs and temporary files of old continuous activities
- Clear deleted experiments**
Permanently delete Experiment Tracking experiments/runs in the 'deleted' lifecycle stage.

- 언제 백업을 해야 하나?
 - 주기적으로 Data_Directory 백업 권장 함
 - 업그레이드 전에도 백업 권장
- Full Backup
 - Dataiku DSS를 stop 하고 DATA_DIR 전체 백업
 - ex) `tar -zcvf backup.tar.gz <DATA_DIR>/`
 - 단점: 단순한 반면 full backup은 DATA_DIR 용량이 큰 경우 효율적이지 않음
- 대안
 - rsyn 를 이용한 incremental backup
 - 파일 시스템 레벨의 백업 툴 예)XFS dump



6. 확장 및 고급 설정

- What?

- 사용자가 확장 기능을 제작할 수 있는 기능 및 툴 세트
- 주로 R 또는 Python 으로 작성 함

- 유용한 링크

- Dataiku 등록 plugins: <https://www.dataiku.com/product/plugins/>
- 공식 문서 : <https://doc.dataiku.com/dss/latest/plugins/index.html>
- 개발자 가이드: <https://academy.dataiku.com/plugin-development>
- Public git 리파지토리: <https://github.com/dataiku/dataiku-contrib>

Plug-in > Store

Dataiku DSS

Search DSS

Plugins (166 in store)

Store Installed Development **ADD PLUGIN**

Search plugin...

Tags (72) - Show less

- ☒ All
- ☐ Connector 35
- ☐ Machine Learning 33
- ☐ Cloud 33
- ☐ NLP 30
- ☐ Visualization 12
- ☐ Enrichment 12
- ☐ Geospatial 12
- ☐ Generative AI 12
- ☐ Computer Vision 10
- ☐ Data Preparation 10
- ☐ API 10
- ☐ Deep Learning 10
- ☐ Google 9
- ☐ Format 9
- ☐ Image 8
- ☐ Open Data 8
- ☐ Productivity 7
- ☐ AWS 7
- ☐ Governance 7
- ☐ Time Series 7
- ☐ AutoML 6
- ☐ Agents 6
- ☐ CRM 5
- ☐ Tools 5
- ☐ Graph 4
- ☐ Microsoft Azure 3

Image annotations to Dataset

By Dataiku (Claire BEHUE)

Plugin that converts image annotations into a dataset compatible with object detection in DSS

Deep Learning Data Preparation [SEE DETAILS](#) [INSTALL](#)

Time Series Preparation

By Dataiku (Du PHAN and Marine SOBAS)

Perform resampling, windowing operations, interval extraction, extrema extraction, and decomposition on time series data (one row per time stamp).

Time Series [SEE DETAILS](#) [INSTALL](#)

AKS clusters

By Dataiku

Interact with or create Microsoft Azure Kubernetes Service clusters

Cloud Microsoft Azure [SEE DETAILS](#) [INSTALL](#)

Amazon Comprehend NLP

By Dataiku (Alex COMBESSIE, Joachim ZENTICI)

Use Amazon Comprehend APIs to perform Language Detection, Sentiment Analysis, Named Entity Recognition and Key Phrase Extraction

AWS Cloud NLP [SEE DETAILS](#) [INSTALL](#)

Amazon Comprehend Medical

By Dataiku (Alex COMBESSIE)

Use Amazon Comprehend Medical APIs to extract Protected Health Information and Medical Named Entities from medical text records

AWS Cloud NLP [SEE DETAILS](#) [INSTALL](#)

Amazon Rekognition

By Dataiku (Alex COMBESSIE, Joachim ZENTICI)

Use Amazon Rekognition APIs to perform Object Detection & Labeling, Text Detection and Unsafe Content Moderation on images

AWS Cloud Image [SEE DETAILS](#) [INSTALL](#)

Azure Cognitive Services - Text Analytics

By Dataiku (Alex COMBESSIE, Joachim ZENTICI)

Dataproc clusters

By Dataiku (Alex COMBESSIE, Joachim ZENTICI)

RELOAD ALL

Plugins are not supported by default.

Plug-in > Installed

Search DSS...

?

A

Plugins (5 installed)

Store

Installed

Development

ADD PLUGIN

Search plugin...

Tags (4)

☒ All

☐ Code-studio 1

☐ Editors 1

☐ Web-apps 1

☐ Charts 1

Support Coverage ?

☐ Supported 0

☐ Tier 2 Support 0

Plugins are not supported by default.

Plugins covered by Dataiku Support or tier 2 support are explicitly indicated.

Name ▲	By	Origin	Version	Description	RELOAD ALL
Builtin macros	Dataiku	Builtin	1.0.0	A default set of macros	
Code Studio built-in blocks	Dataiku	Builtin	1.0.0	A collection of useful editors and frameworks to use as blocks in your Code Studios : JupyterLab, RStudio, VS Code, Streamlit.	
ColorBrewer palettes	Dataiku	Builtin	0.0.1	Make ColorBrewer color palettes usable in charts. This plugin includes color specifications and designs developed by Cynthia Brewer (http://colorbrewer.org/).	
Kubernetes metrics integrations	Dataiku	Builtin	1.0.0	This plugin contains pod enhancers in order to provide deep integration with various metrics reporting capabilities in Kubernetes and cloud providers	
Local R Development environment setup tools	Dataiku	Builtin	1.0.0		

Plug-in > Development

The screenshot displays the Dataiku 'Development' tab for plugins. The main table lists two plugins: 'Plugin test' and 'Poscodx test', both by 'dohyun' with version '0.0.1'. Two red arrows point to these entries. An inset shows the 'Poscodx test' plugin details page, which includes a 'Code environment' section and a message stating 'This plugin has no components.' with a '+ CREATE YOUR FIRST COMPONENT' button.

Name	By	Version	Description
Plugin test	dohyun	0.0.1	
Poscodx test	dohyun	0.0.1	

Poscodx test

Author: dohyun
Version: 0.0.1
License: Apache Software License
Location: /home/dataiku/dss/plugins/dev/poscodx-test

Code environment

This plugin requires a dedicated code environment.
Plugin currently uses env plugin_poscodx-test_managed_1
[CHANGE] [DISSOCIATE]

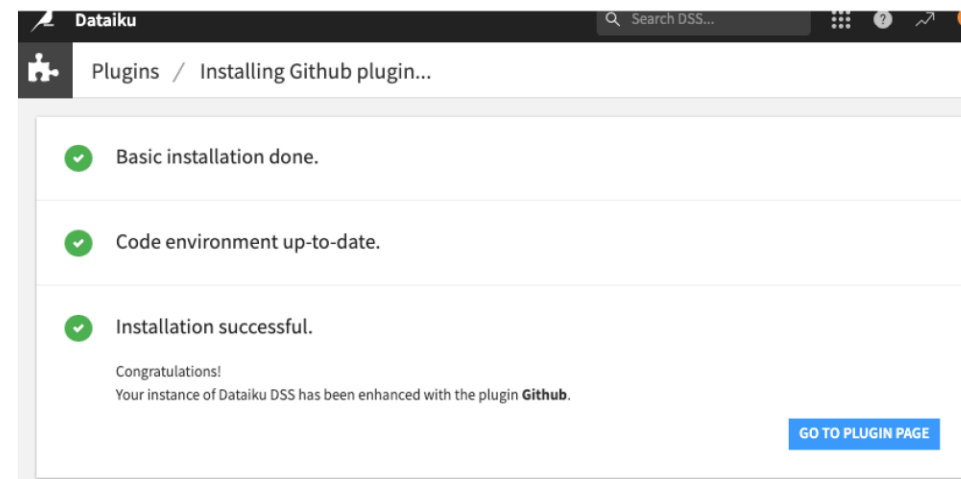
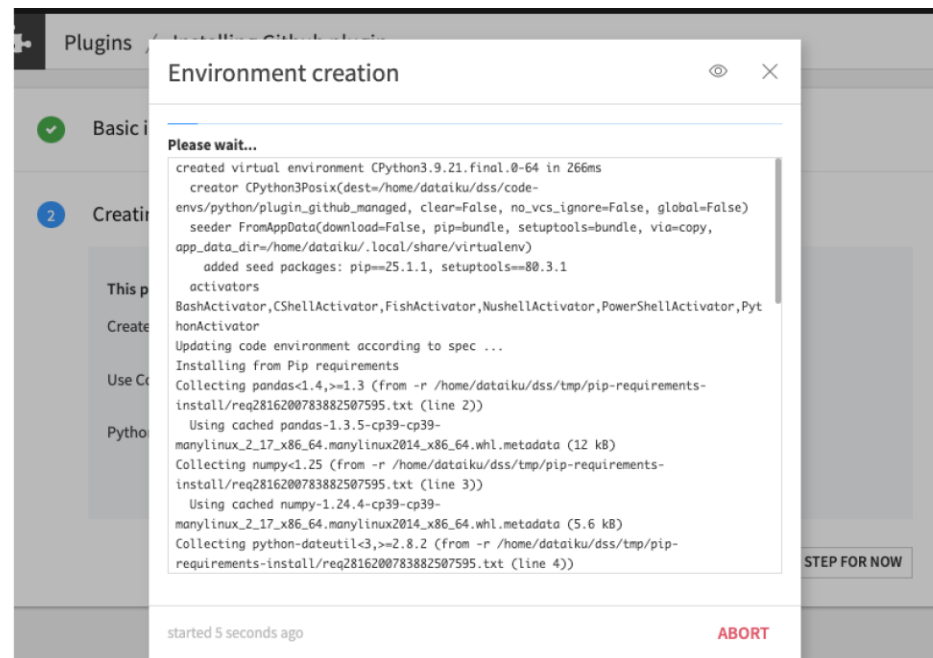
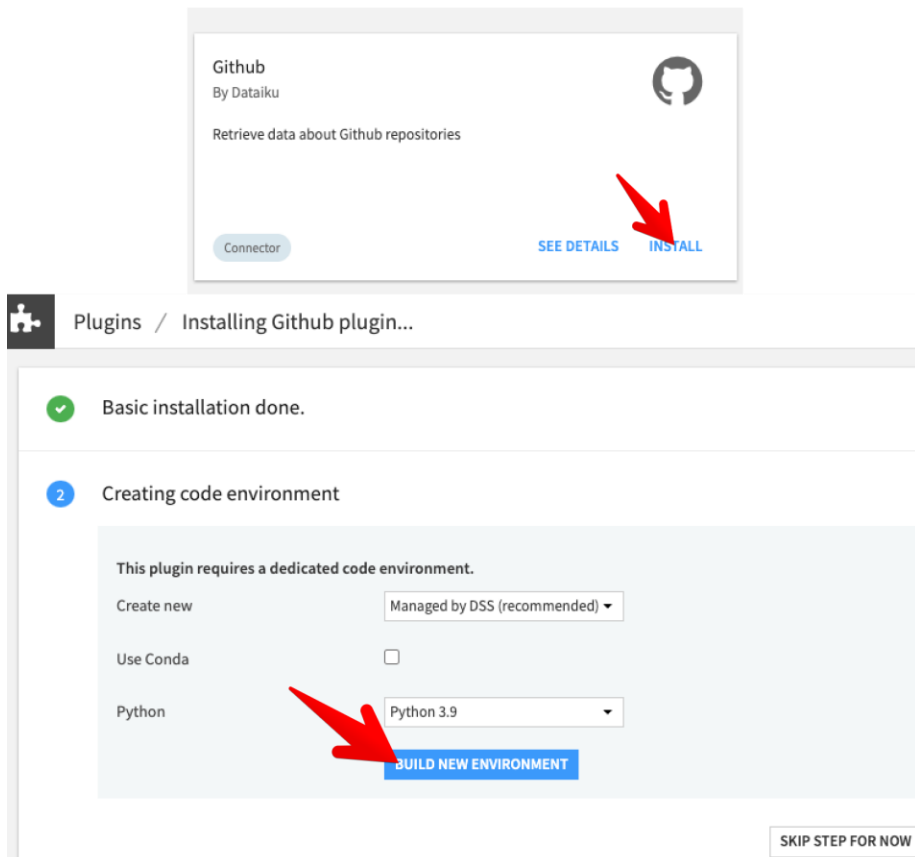
This plugin has no components.

A plugin is made of a number of components, each component being a specific new kind of object in DSS.

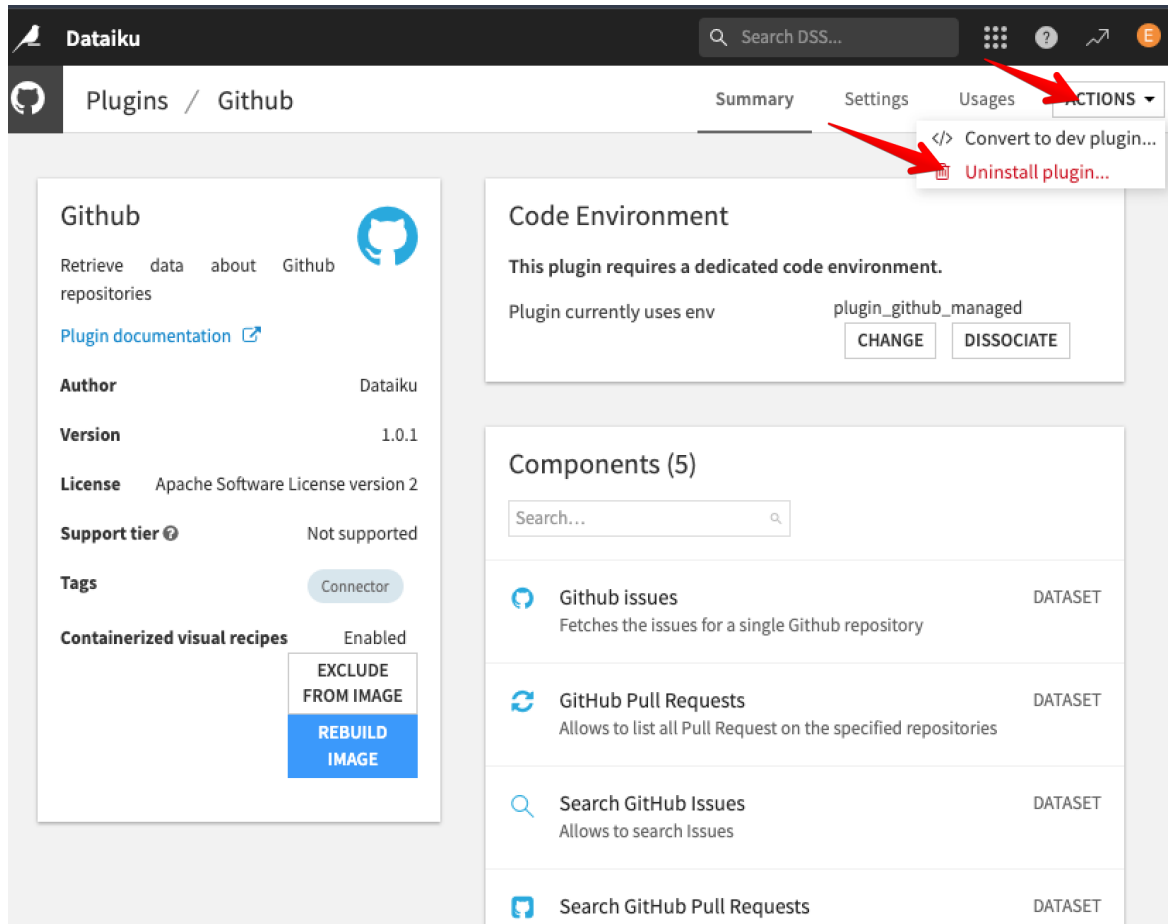
[Read the documentation](#) to learn more

[+ CREATE YOUR FIRST COMPONENT](#)

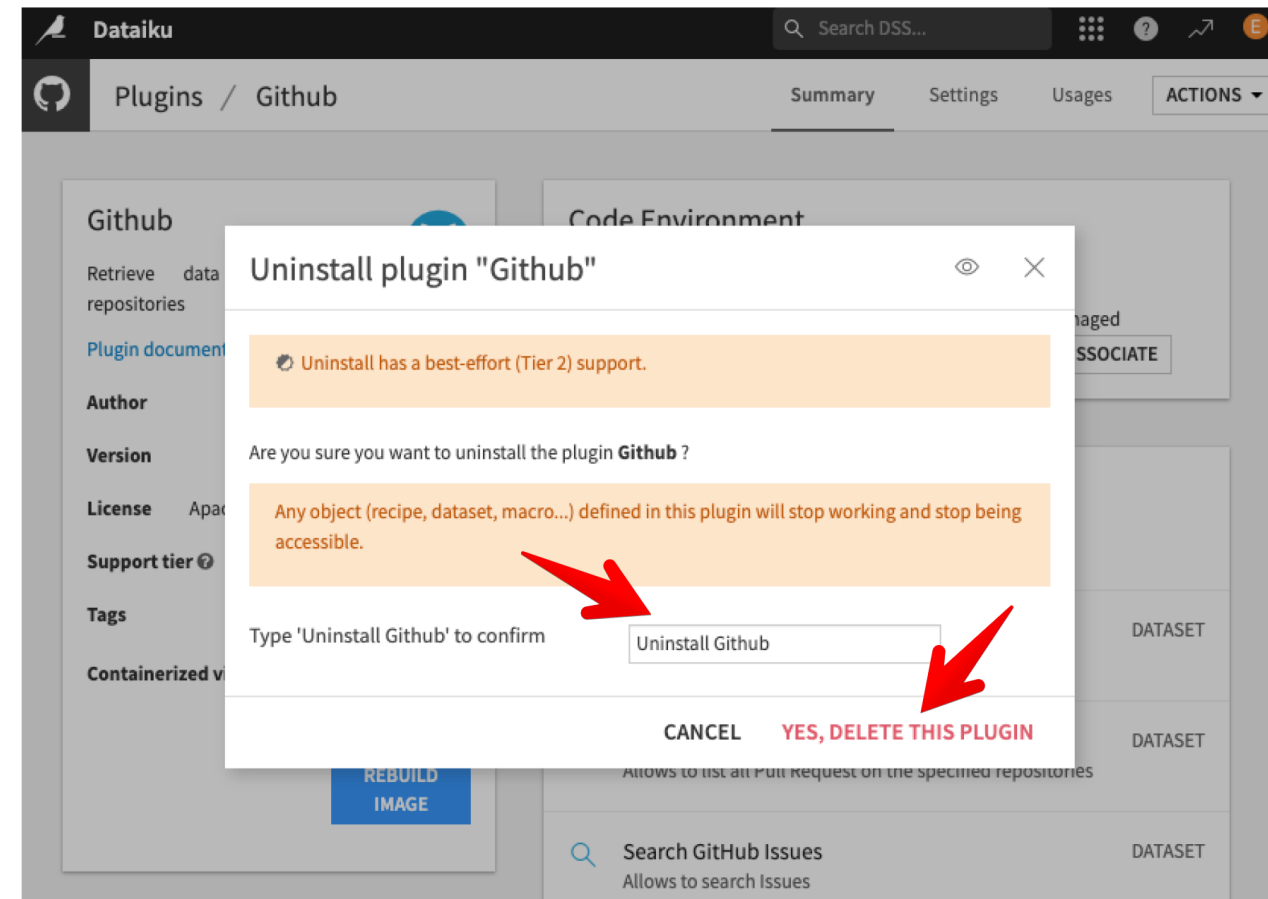
Lab > GitHub 플러그인 설치



Lab > GitHub 플러그인 삭제



The screenshot shows the Dataiku interface for the GitHub plugin. The top navigation bar includes 'Summary', 'Settings', 'Usages', and 'ACTIONS'. The 'ACTIONS' menu is open, showing options: '</> Convert to dev plugin...' and 'Uninstall plugin...'. The main content area is divided into two sections: 'Github' on the left and 'Code Environment' on the right. The 'Github' section includes details like 'Retrieve data about Github repositories', 'Plugin documentation', 'Author: Dataiku', 'Version: 1.0.1', 'License: Apache Software License version 2', 'Support tier: Not supported', 'Tags: Connector', and 'Containerized visual recipes: Enabled'. The 'Code Environment' section states 'This plugin requires a dedicated code environment.' and 'Plugin currently uses env: plugin_github_managed', with 'CHANGE' and 'DISSOCIATE' buttons. Below these is a 'Components (5)' section listing various GitHub-related datasets.



The screenshot shows the same Dataiku interface, but with the 'Uninstall plugin "Github"' dialog box open. The dialog box contains the following text: 'Uninstall has a best-effort (Tier 2) support.', 'Are you sure you want to uninstall the plugin Github?', and 'Any object (recipe, dataset, macro...) defined in this plugin will stop working and stop being accessible.' Below this is a text input field with 'Uninstall Github' and a 'Type \'Uninstall Github\' to confirm' label. At the bottom are 'CANCEL' and 'YES, DELETE THIS PLUGIN' buttons. Red arrows point to the 'Uninstall plugin...' option in the first screenshot, the 'Uninstall Github' text in the input field, and the 'YES, DELETE THIS PLUGIN' button in this screenshot.

Code Environments

- What

- 프로젝트/레시피/플로그인이 사용하는 독립 실행환경 (Python/R)
- 각 Code Env는 개별적으로 패키지 세트를 유지 함
- 종류: DSS Managed와 사용자 Managed
- 이동성: Bundle/Deployer 배포 시 환경 스펙을 함께 이동 함

- So-What?

- 재현성 확보: 팀.환경이 달라도 같은 코드 = 같은 결과
- 리스크 감소: 라이브러리 업데이트/충돌로 인한 운영 장애 최소화
- 성능/하드웨어 활용: GPU/CUDA·네이티브 의존성을 환경별로 최적화(예 PyTorch/TensorFlow)

Lab. Code Environment 생성

New Python env

Deployment typeManaged by DSS (recommended)

Nametest

PythonPython 3.9 - Not available

- Python 3.5 - Not available
- Python 3.6 - Not available
- Python 3.7 - Not available
- Python 3.8 - Not available
- Python 3.9 - Not available
- Python 3.10
- Python 3.11 - Not available
- Python 3.12 - Not available
- Custom (lookup in PATH)

Conda

Mandatory packages

Jupyter

CREATE

Dataiku DSS

DSS s... License Connections Security Clusters Code Studios Code Envs

test

General Packages to install **Currently installed packages** Permissions Containerized execution Resources Logs Usages

SAVED! UPDATE

☒ Update all packages
Updates packages in the virtual environment so that the installed packages match the list and versions of the packages specified by the user.

☒ Run resources init script
Clears the code environment and rebuilds it from scratch

☐ Rebuild env

Currently installed packages

Installed packages (Pip)

These are the packages that are **currently** installed in the code env. Note that only the list in "Packages to install" is authoritative as to which packages will be installed.
Packages that are not in "Packages to install" may not be installed when rebuilding this env or in containers. This list only reflects the "local" code env, not what is available in containers.

asttokens==3.0.0
backcall==0.2.0
certifi==2025.10.5
charset-normalizer==3.4.4
comm==0.2.3
debugpy==1.8.17
decorator==5.1.1
executing==2.2.1
idna==3.11
ipykernel==6.23.3
ipython==8.12.3
ipython-genutils==0.2.0
jedi==0.19.2
jupyter-client==6.1.12
jupyter_core==4.12.0
matplotlib-inline==0.2.1
nest-asyncio==1.6.0
numpy==1.26.4
packaging==25.0
pandas==2.2.3
parso==0.8.5

Lab. Code Environment 생성 > 패키지 추가

test

General

Packages to install

Currently installed packages

Permissions

Containerized execution

Resources

Logs

Usages

SAVE SAVE AND UPDATE

be able to use Dataiku APIs without this)

Install Jupyter support ☒ Install support for Jupyter notebooks

Requested packages (Pip)

Add here the packages that you want to be available in the environment. **Do not remove packages once they are installed.** Only packages mentioned here are guaranteed to be installed.

ADD SETS OF PACKAGES

```
1 # Redis 패키지 추가
2 redis>=5.0,<6.0
3 # 초고속 JSON 직렬화(대용량 메시지)
4 orjson>=3.9
```

```
# Redis 패키지 추가
redis>=5.0,<6.0
# 초고속 JSON 직렬화(대용량 메시지)
orjson>=3.9
```


Add additional packages

Some actions in DSS require to use a code-env with certain packages that are not shipped in DSS built-in code-env.

Required packages for

Native access to streams (Kafka, HTTP SSE ▼)

The following packages will be added to the requested packages.

```
pykafka==2.8.0
sseclient==0.0.26
```

Lab. Code Environment 생성 > Permissions

test

General

Packages to install

Currently installed packages

Permissions

Containerized execution

Resources

Logs

Usages

SAVE

SAVE AND UPDATE

☒ Update all packages

Updates packages in the virtual environment so that the installed packages match the list and versions of the packages specified by the user.

☒ Run resources init script

☐ Rebuild env

Clears the code environment and rebuilds it from scratch.

Permissions

Owner

So Jinyoung

Usable by all☒

Group name	Use	Update settings & packages	Admin
AI_SECTION	☒	☒	☐
data_team	+ GRANT ACCESS TO GROUP		
Filter... administrators data_team readers			

Generate Prepare Steps & Generate Recipe

The screenshot shows the Dataiku interface with a data flow diagram and a dataset preview. The flow starts with two datasets, 'car_data_1_Sheet1' and 'car_data_2_Sheet1', which are joined into 'car_data_1_Sheet1_joined'. This is then prepared into 'car_data_1_Sheet1_joined_prepared2'. The flow then splits into two paths, one for '2024' and one for '2025', which are then joined back together. The dataset preview at the bottom shows the following data:

차량ID	제조사 및 차종	연도	가격	변속기	사용거리
11675	독일/merc/GLC Cl...	2024	41990	Automatic	4569
15210	독일/audi/A6	2024	37999	Automatic	2234
41841	독일/merc/E Class	2024	24749	Automatic	15210
12243	독일/audi/TT	2024	25495	Semi-Auto	10018

The screenshot shows the 'Generate recipe from the selected dataset' panel in Dataiku. The panel displays a recipe generated from the selected dataset 'car_data_1_Sheet1'. The recipe is titled 'Generate recipe from the selected dataset' and contains the following steps:

- Reformulation**: Divide the 가격 column by 1000 and store the result in a new column called 가격K.
- Reasoning**: I will use a prepare recipe to create a new column by dividing the existing 가격 column by 1000.

The panel also includes a warning: 'AI-generated results may be incorrect. Please exercise caution.' and buttons for 'EDIT PROMPT' and 'CREATE RECIPE'.

Lab. Settings> AI Services

The screenshot shows the Dataiku DSS interface. The top navigation bar includes a search bar and icons for help, refresh, and user profile. The main navigation menu on the left lists various settings categories: DSS settings, License, Connections, Security, Clusters, Code Studios, Code Envs, Monitoring, Maintenance, and Settings. The 'Settings' menu item is highlighted with a red arrow. The left sidebar contains a list of settings categories: ENGINES & FLOW, COMPUTE & SCALING, SECURITY & AUDIT, and OTHER. The 'AI Services' link under the 'OTHER' category is highlighted with a red arrow. The main content area displays the 'AI Services' settings page, which includes a description of the features and a section titled 'Generate Prepare Steps & Generate Recipe'.

Dataiku DSS

Search DSS...

DSS settings License Connections Security Clusters Code Studios Code Envs Monitoring Maintenance **Settings**

ENGINES & FLOW

- Engines & connections
- Global Variables

COMPUTE & SCALING

- Containerized execution
- Spark
- Resources control
- Metastore catalogs
- Hadoop

SECURITY & AUDIT

- User login & provisioning
- Git security
- Auditing
- Other security settings

OTHER

- Deployer
- Dataiku Govern
- LLM Mesh
- AI Services**

AI Services

These are the settings for the features and capabilities in Dataiku allowing you to leverage AI Services to increase analytics productivity.

Generate Prepare Steps & Generate Recipe

Generate Steps allows you to explain in natural language what you want to do in a Prepare Recipe. Generate Steps then automatically generates Prepare processor steps.

Generate Recipe allows you to explain in natural language what you want to do in a flow to create a new recipe. Generate Recipe then automatically generates the corresponding visual recipe.

To enable this feature, you must first [agree to the Dataiku AI Services Terms of Use](#)

AI Generate SQL

AI Generate SQL allows you transform natural language into accurate SQL queries. It simplifies queries tailored to your needs in seconds.

To enable this feature, you must first [agree to the Dataiku AI Services Terms of Use](#)

AI Explain

AI Explain provides explanations of what your Flow does.

To enable this feature, you must first [agree to the Dataiku AI Services Terms of Use](#)

The modal dialog titled 'AI Services Terms of Use' is displayed. It contains the following text: 'Usage of the Dataiku AI Services requires agreeing to the [Dataiku AI Services Terms of Use](#)'. Below this, there is a checkbox labeled 'Agree to the terms' which is checked. To the right of the checkbox, the text reads: 'I have read and agree to the Dataiku AI Services Terms of Use'. At the bottom right of the modal, there are two buttons: 'CANCEL' and 'OK'. A red arrow points to the 'OK' button.

AI Services Terms of Use

Usage of the Dataiku AI Services requires agreeing to the [Dataiku AI Services Terms of Use](#)

Agree to the terms ☒ I have read and agree to the Dataiku AI Services Terms of Use

CANCEL OK

Lab. Settings> AI Services

Global Variables

COMPUTE & SCALING

Containerized execution

Spark

Resources control

Metastore catalogs

Hadoop

SECURITY & AUDIT

User login & provisioning

Git security

Auditing

Other security settings

OTHER

Deployer

Dataiku Govern

LLM Mesh

AI Services

Access & requests

Misc.

 SAVE

Generate Prepare Steps & Generate Recipe

Generate Steps allows you to explain in natural language what you want to do in a Prepare Recipe. Generate Steps then automatically generates Prepare processor steps.

Generate Recipe allows you to explain in natural language what you want to do in a flow to create a new recipe. Generate Recipe automatically generates the corresponding visual recipe.

Usage of this feature is governed by the [Dataiku AI Services Terms of Use](#). Metadata about processed dataset (and, optionally,) sent to Dataiku and third-party services

Enable Generate Steps & Recipe ☒

Send sample values ☒ Sends sample values from each column. Strongly improves quality of answers. Sample data is not kept by Dataiku nor

Diagnostic data ☒ Help Dataiku improve Generate Steps & Generate Recipe. Allows Dataiku to retain metadata about processed dataset and q retained.

AI Generate SQL

AI Generate SQL allows you transform natural language into accurate SQL queries. It simplifies database interactions by generating queries tailored to your needs in seconds.

Usage of this feature is governed by the [Dataiku AI Services Terms of Use](#). Metadata about processed dataset is sent to Dataiku services

Enable AI Generate SQL ☐

AI Explain

AI Explain provides explanations of what your Flow does.



수고 하셨습니다.